

Optimization

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Part I

Introduction

Topics in the course

- Introduction and generalities about optimization
- Notions of convexity
 - ▶ Convex sets and functions
 - ▶ Separation theorem
- Optimization problems
 - ▶ Convex optimization problems
 - ▶ Optimality conditions
- Duality
 - ▶ Lagrange duality
 - ▶ Conjugate function and Fenchel duality
 - ▶ Karush-Kuhn-Tucker optimality conditions
- Algorithms
 - ▶ Notions on unconstrained optimization (gradient, Newton)
 - ▶ Notions on constrained optimization (interior points)
 - ▶ Basic introduction to proximal methods

Optimization softwares

Many free and commercial softwares exist for optimization:

- optimization solvers: SeDuMi, SDPT3, CPLEX, Gurobi, Mosek, ...
- high level modelling languages and parsers: CVX, YALMIP, ...

but many algorithms are not that complicated and can be programmed (e.g. with Matlab/Scientific Python/Julia)!

Useful references

Convex optimization:

- Boyd and Vandenberghe, *Convex Optimization* (Cambridge University Press)
- <http://stanford.edu/~boyd/>
- Borwein and Lewis, *Convex Analysis and Nonlinear Optimization, Theory and Examples* (Canadian Mathematical Society)

Proximal algorithms:

- N. Parikh and S. Boyd, *Proximal Algorithms* (Foundations and Trends in Optimization, 1(3):123-231, 2014)
- S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, *Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers* (Foundations and Trends in Machine Learning, 3(1):1-122, 2011.)

Optimization problems

Unconstrained optimization problem

Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, find $x^* \in \mathbb{R}^n$ such that:

$$\forall x \in \mathbb{R}^n : \quad f(x^*) \leq f(x)$$

Constrained optimization problem

Given functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ for $i = 1, \dots, m$, find x^* such that:

$$g_i(x^*) \leq 0, i = 1, \dots, m$$

$$f(x^*) \leq f(x), \quad \forall x \in \mathbb{R}^n \text{ such that } g_i(x) \leq 0, i = 1, \dots, m$$

Discrete optimization (not covered in this course):

f and g_i are functions $\mathcal{D} \rightarrow \mathbb{R}$ with:

- $\mathcal{D}$ finite : combinatorial optimization problem
- $\mathcal{D} = \mathbb{Z}$: integer programming

Optimization problem

$$\begin{cases} \min. f(x) \\ \text{s.t. } g_i(x) \leq 0, \quad i = 1, \dots, m \end{cases}$$

- $x = (x_1, \dots, x_n)^\top$: optimization variables
- $f : \mathbb{R}^n \rightarrow \mathbb{R}$: objective function
- $g_i : \mathbb{R}^n \rightarrow \mathbb{R}, i = 1, \dots, m$: constraint functions

Optimal value: $p^\star := \inf\{f(x) \mid g_i(x) \leq 0, \text{ for } i = 1, \dots, m\}$

Optimal solution: $x^\star$ satisfies $g_i(x^\star) \leq 0, i = 1, \dots, m$ and:

$$f(x^\star) \leq f(x) \text{ for all } x \text{ that satisfy } g_i(x) \leq 0, i = 1, \dots, m.$$

Examples

Portfolio optimization

- variables: amounts invested in different assets
- constraints: budget, max./min. investment per asset, minimum return
- objective: overall risk or return variance

Data fitting

- variables: model parameters
- constraints: prior information, parameter limits
- objective: measure of misfit or prediction error

Signal restoration

- variables: signal values
- constraints: prior informations, value limits
- objective: data fit + regularization

Example: (linear) classification

- Training data $(a_i, c_i)_{i=1, \dots, m}$ where for any $i = 1, \dots, m$:
 - ▶ $a_i \in \mathbb{R}^n$: features,
 - ▶ $c_i \in \{+1, -1\}$: category.
- Classify new data $f \in \mathbb{R}^n$ in the two classes.
 Linear classifier $\hat{c} = \text{sign}(x^\top a)$: find weight vector x
- Associated optimization problem with ℓ_2 regularization:

$$\min_x \sum_{i=1}^m \varphi\left(-c_i(x^\top a_i)\right) + \gamma \|x\|_2 \quad (\gamma = \text{const.} > 0)$$

where cost function $\varphi(z)$ can be:

- ▶ $\varphi(z) = \mathbb{1}(z \geq 0)$
- ▶ $\varphi(z) = \log(1 + e^{-z})$ (logistic regression)
- ▶ $\varphi(z) = [1 - z]_+$ (support vector machine)
- ▶ $\varphi(z) = e^z$

Convex or non convex (1/2)

General optimization problem:

- very difficult to solve (if nonconvex)
- methods involve some compromise, e.g.:
 - ▶ **local** optimization method (*nonlinear programming*): not always finding the solution
 - ▶ **global** optimization: very long computation time, worst case complexity grows exponentially with problem size

~> These algorithms are often based on solving convex subproblems

Convex or non convex (2/2)

Convex optimization problems

- Objective and constraint functions are convex
- Includes as special cases: least squares, linear programming,...
- Can be **solved efficiently and reliably**; (Convex optimization is «almost a technology»:
 - ▶ iterative algorithms (but generally no analytical solutions)
 - ▶ computation time (roughly) proportional to $\max\{n^3, n^2m, F\}$ where F is cost of evaluating f, g_i ($i = 1, \dots, m$) and their first+second derivatives
- Many problems can be solved via convex optimization:
 - ▶ often difficult to recognize
 - ▶ many tricks for transforming problems

Gradient vector (scalar valued function)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$.

- **Gradient** (column) vector $\nabla f(x)$:

$$[\nabla f(x)]_i = \frac{\partial f(x)}{\partial x_i}$$

First-order approximation of f near $\bar{x}$:

$$\begin{aligned} f(x) &= f(\bar{x}) + \nabla f(\bar{x})^\top (x - \bar{x}) \\ &\quad + o(\|x - \bar{x}\|) \end{aligned}$$

- Ex:

$$f(x) = a^\top x$$

$$\nabla f(x) = a$$

$$g(x) = x^\top M x$$

$$\nabla g(x) = (M + M^\top)x$$

$$= 2Mx \text{ if } M \text{ symmetric.}$$

Hessian matrix (scalar valued function)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$.

- **Hessian matrix** $\nabla^2 f(x)$

$$[\nabla^2 f(x)]_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}$$

Second-order approximation of f near $\bar{x}$:

$$\begin{aligned} f(x) = f(\bar{x}) + \nabla f(\bar{x})^\top (x - \bar{x}) + \frac{1}{2} (x - \bar{x})^\top \nabla^2 f(\bar{x}) (x - \bar{x}) \\ + o(\|x - \bar{x}\|^2) \end{aligned}$$

- Ex:

$$f(x) = a^\top x$$

$$\nabla^2 f(x) = 0$$

$$g(x) = x^\top M x$$

$$\nabla^2 g(x) = M + M^\top$$

$= 2M$ if M symmetric.

Differential, Jacobian, Gradient (vector valued function)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$

- **Differential** is the unique linear application $u \mapsto df(\bar{x}; u)$ such that:

$$f(\bar{x} + u) = f(\bar{x}) + df(\bar{x}; u) + o(\|u\|)$$

$df(\bar{x}; u)$ = differential of f at $\bar{x}$ with increment u

- **Jacobian matrix** contains partial derivatives $[Df(\bar{x})]_{ij} = \frac{\partial f_i}{\partial x_j}(\bar{x})$ and:

$$df(\bar{x}; u) = Df(\bar{x})u$$

If $m = n$, the *Jacobian* is $\det(Df(\bar{x}))$.

- **Gradient** $\nabla f(\bar{x}) = Df(\bar{x})^\top$ and hence $df(\bar{x}; u) = \nabla f(\bar{x})^\top u$

$\rightsquigarrow Df(\bar{x})$ is $m \times n$ $\nabla f(\bar{x})$ is $n \times m$

$\rightsquigarrow$ Ex: For linear function $g(x) = Ax$ (with $A \in \mathbb{R}^{m \times n}$): $\nabla g(\bar{x}) = A^\top$

Differential and composition

Let $g : \mathbb{R}^d \rightarrow \mathbb{R}^n$, $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and consider $h(x) = f(g(x))$.

Chain rule

$$d(f \circ g)(\bar{x}; \bullet) = df(g(\bar{x}); dg(\bar{x}; \bullet))$$

Indeed, write $h(\bar{x} + u)$ using:

$$\begin{aligned} g(\bar{x} + u) &= \underbrace{g(\bar{x})}_{\bar{y}} + \underbrace{dg(\bar{x}; u)}_v + o(\|u\|) \\ f(\bar{y} + v) &= f(\bar{y}) + df(\bar{y}; v) + o(\|v\|) \end{aligned}$$

→ For gradients: $\nabla(f \circ g)(\bar{x}) = \nabla g(\bar{x}) \nabla f(g(\bar{x}))$

• Ex: linear (pre-)composition $h(x) = f(Ax)$:

$$dh(\bar{x}; u) = df(A\bar{x}; Au) \quad \nabla h(\bar{x}) = A^\top \nabla f(A\bar{x})$$

Gradient in different metrics (scalar valued function)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ ($m = 1$)

$$f(\bar{x} + u) = f(\bar{x}) + \underbrace{df(\bar{x}; u)}_{\nabla f(\bar{x})^\top u} + o(\|u\|)$$

- **Gradient** $\nabla f(\bar{x})$ given by column vector $\left(\frac{\partial f}{\partial x_i}(\bar{x})\right)_{i=1}^n$ is such that

$$df(\bar{x}; u) = \langle \nabla f(\bar{x}), u \rangle$$

with **usual inner product** $\langle x, y \rangle = x^\top y$. in $\mathbb{R}^n$.

- Differential does not depend on inner product.

→ Gradient in a different metric $\langle \cdot, \cdot \rangle_W$ (e.g. $\langle x, y \rangle_W = x^\top W y$) defined by:

$$df(\bar{x}; u) = \langle \nabla_W f(\bar{x}), u \rangle_W$$

Domain and extended-value function

Let f be a function $\mathbf{E} \rightarrow \mathbb{R}$ (often, $\mathbf{E} = \mathbb{R}^n$).

Domain: $\text{dom } f = \{x \in \mathbf{E} \mid f(x) \text{ exists}\}$ ($\text{dom } f \subset \mathbf{E}$)

If $f : \text{dom } f \rightarrow \mathbb{R}$, we use the extended-value extension of f :

$$f : \mathbf{E} \rightarrow \mathbb{R} \cup \{+\infty\}$$
$$x \mapsto \begin{cases} f(x) & \text{if } x \in \text{dom } f \\ +\infty & \text{if } x \notin \text{dom } f \end{cases}$$

- ▶ Often simplifies the notation and provides a unifying view.
- ▶ $\text{dom } f = \{x \in \mathbf{E} \mid f(x) < \infty\}$
 - If $\text{dom } f \neq \emptyset$, the function is said **proper**

Extended-value functions

Examples

- **Log-barrier** $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by:

$$f(x) = \begin{cases} -\log(-x) & \text{if } x < 0, \\ +\infty & \text{if } x \geq 0. \end{cases}$$

$$\text{dom } f = \mathbb{R}_{--}$$

- **Indicator** function of a given set $C \subset \mathbf{E}$:

$$\iota_C(x) = \begin{cases} 0 & \text{if } x \in C, \\ +\infty & \text{otherwise.} \end{cases}$$

$$\text{dom } \iota_C = C$$

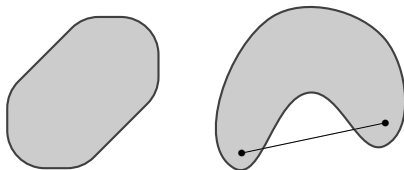
Part II

Convexity, convex optimization

Convex set

Convex set: contains line segment between any two points in the set

$$x, y \in C, 0 \leq \theta \leq 1 \Rightarrow \theta x + (1 - \theta)y \in C$$

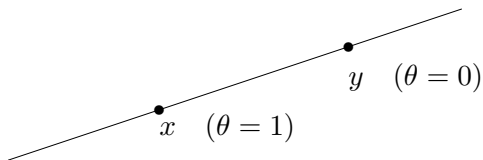


- Points of the form $\theta x + (1 - \theta)y$ with $0 \leq \theta \leq 1$ correspond to the line segment between x and y .

Affine set

Affine set: the line through any two points in the set is contained in the set

$$x, y \in C, \theta \in \mathbb{R} \Rightarrow \theta x + (1 - \theta)y \in C$$



- Points of the form $\theta x + (1 - \theta)y$ with $\theta \in \mathbb{R}$ correspond to the line through x and y .

Affine and convex hull

Affine hull of set C = all affine combinations of points in C

$$\text{aff } C = \{\theta_1 x_1 + \cdots + \theta_k x_k \mid x_i \in C, \theta_1 + \cdots + \theta_k = 1\}$$

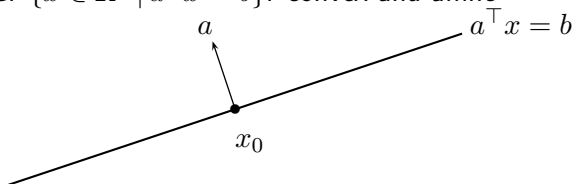
Convex hull of set C = all convex combinations of points in C

$$\text{conv } C = \{\theta_1 x_1 + \cdots + \theta_k x_k \mid x_i \in C, \theta_i \geq 0, \theta_1 + \cdots + \theta_k = 1\}$$

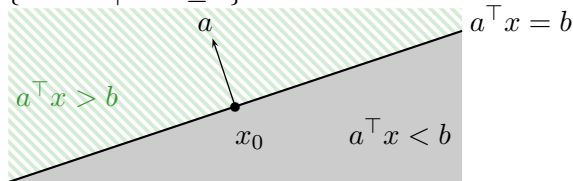
Hyperplanes and halfspaces in $\mathbb{R}^n$

Let $a \in \mathbb{R}^n, a \neq 0$ and $b \in \mathbb{R}$:

- **Hyperplane:** $\{x \in \mathbb{R}^n \mid a^\top x = b\}$: convex and affine



- **Halfspace:** $\{x \in \mathbb{R}^n \mid a^\top x \leq b\}$: convex but not affine



- a is the normal vector
- The hyperplane separates the whole space $\mathbb{R}^n$ in two halfspaces

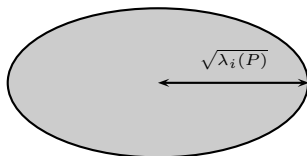
Balls and ellipsoids

Euclidian ball: convex, non affine

$$\begin{aligned} B(\bar{x}, r) &= \{x \mid \|x - \bar{x}\|_2 \leq r\} = \{x \mid (x - \bar{x})^\top (x - \bar{x}) \leq r^2\} \\ &= \{\bar{x} + ru \mid \|u\|_2 \leq 1\} \end{aligned}$$

Ellipsoid: convex, non affine

$$\mathcal{E} = \{x \mid (x - \bar{x})^\top P^{-1}(x - \bar{x}) \leq 1\} \quad \text{where } P \in \mathbb{S}_{++}^n$$



With $A = P^{1/2}$, other representation: $\mathcal{E} = \{\bar{x} + Au \mid \|u\|_2 \leq 1\}$

Operations that preserve convexity (1/3)

Intersection : the intersection of any number of convex sets is convex

Ex:

- **Polyhedra**: intersection of a finite number of hyperplanes/halfspaces
 - ▶ $\mathcal{P} = \{x \mid a_j^\top x \leq b_j, j = 1, \dots, m, c_i^\top x = d_i, i = 1, \dots, p\}$
 - ▶ Simplex $\{\theta_0 v_0 + \dots + \theta_k v_k \mid \theta \succeq 0, \mathbf{1}^\top \theta = 1\}$
 $(v_0, \dots, v_k \text{ affinely independent})$
- Intersection of halfspaces:

$$\{x \in \mathbb{R}^m \mid \sum_{k=1}^m x_k \cos kt \leq 1, \forall t \in [-\pi/3, \pi/3]\}$$
- Positive semidefinite matrices: $\mathbb{S}_+^n = \bigcap_{x \neq 0} \{M \in \mathbb{S}^n \mid x^\top M x \geq 0\}$

Convex hull of a set S : intersection of all convex sets containing S .

Operations that preserve convexity (2/3)

Affine transformation: the image and inverse image of a convex set under an affine function is convex.

Ex:

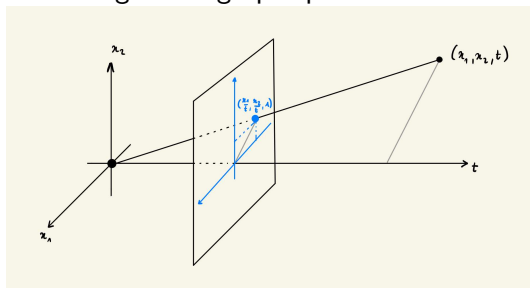
- Scaling, translation, projection.
- Sum $S_1 + S_2 = \{x + y ; x \in S_1, y \in S_2\}$
- Partial sum $\{(x, y_1 + y_2) ; (x, y_1) \in S_1, (x, y_2) \in S_2\}$
- Polyhedron (inverse image of nonnegative orthant)
- Ellipsoid (image/inverse image of the unit Euclidian ball)
- Solution set of a Linear Matrix Inequality (LMI):
 $\{x \in \mathbb{R}^n \mid x_1 A_1 + \dots + x_n A_n \preceq B\}$
where $B, A_1, \dots, A_n$ are given in $\mathbb{S}^p$

Operations that preserve convexity (3/3)

Perspective function

$$P(x, t) = \frac{x}{t} \text{ where } P : \mathbb{R}^n \times \mathbb{R}_{++} \rightarrow \mathbb{R}^n$$

→ image and inverse image through perspective remains convex.



Linear-fractional $f(x) = \frac{Ax+b}{c^\top x+d}$ with $\text{dom } f = \{x \mid c^\top x + d\} > 0 \rightarrow$ preserve convexity (as a composition of affine and perspective functions).

Relative interior

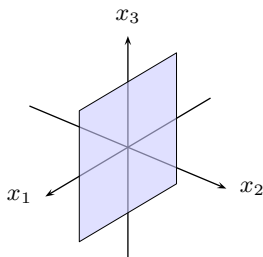
interior of a set C :

$$\text{int } C = \{x \in C \mid B(x, \varepsilon) \subset C \text{ for sufficiently small } \varepsilon\}$$

relative interior of a set C = interior of C relative to its affine hull:

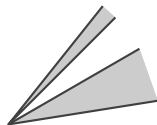
$$\text{relint } C = \{x \in C \mid B(x, \varepsilon) \cap \text{aff } C \subseteq C \text{ for sufficiently small } \varepsilon\}$$

Ex: if $C = \{x \in \mathbb{R}^3 \mid -1 \leq x_1 \leq 1 ; x_2 = 0 ; -1 \leq x_3 \leq 1\}$ then $\text{int } C = \emptyset$ whereas $\text{relint } C = \{x \in \mathbb{R}^3 \mid -1 < x_1 < 1 ; x_2 = 0 ; -1 < x_3 < 1\}$

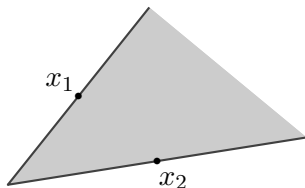


Cones

Cone C : for every $x \in C$ and $\theta \geq 0$, we have $\theta x \in C$



Convex cone C : for every $x_1, x_2 \in C$ and $\theta_1, \theta_2 \geq 0$, we have $\theta_1 x_1 + \theta_2 x_2 \in C$



- Conic hull of a set C :

$$\{\theta_1 x_1 + \cdots + \theta_k x_k \mid x_i \in C, \theta_i \geq 0, i = 1, \dots, k\}$$

Examples of cones

- **Nonnegative orthant** $\mathbb{R}_+^n = \{x \in \mathbb{R}^n \mid x_i \geq 0, i = 1, \dots, n\}$
- **Positive semidefinite matrices**

$$\mathbb{S}_+^n = \{M \in \mathbb{S}^n \mid M \succeq 0\} = \{M \in \mathbb{S}^n \mid x^\top M x \geq 0, \forall x \in \mathbb{R}^n\}$$

where $\mathbb{S}^n$ is the set of symmetric matrices.

- **Norm cone**

$$\{(x, t) \in \mathbb{R}^{n+1} \mid \|x\| \leq t\}$$

When $\|\cdot\| = \|\cdot\|_2$, also called quadratic / second-order / Lorentz cone

- **Cone of positive polynomials**

$$K = \{p \in \mathbb{R}^n \mid p_1 + p_2 t + \dots + p_n t^{n-1} \geq 0, \forall t \in [0, 1]\}$$

Normal cone

Normal cone to a convex set C at $\bar{x} \in C$:

$$\mathcal{N}_C(\bar{x}) = \{d \in \mathbf{E} \mid \langle d, x - \bar{x} \rangle \leq 0, \forall x \in C\}$$

when $\mathbf{E} = \mathbb{R}^n$, simplifies to:

$$\mathcal{N}_C(\bar{x}) = \{d \in \mathbb{R}^n \mid d^\top (x - \bar{x}) \leq 0, \forall x \in C\}$$

Proper cones, generalized inequalities

Proper cone:

- convex
- closed
- solid (*i.e.* nonempty interior)
- pointed (*i.e.* contains no line: $x \in K, -x \in K \Rightarrow x = 0$)

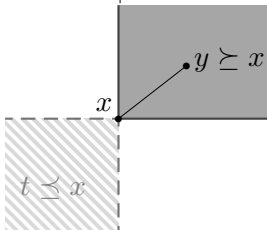
Generalized inequalities w.r.t. proper cone K :

$$x \preceq_K y \Leftrightarrow y - x \in K$$

$$x \prec_K y \Leftrightarrow y - x \in \text{int } K \quad (\text{interior of } K)$$

Examples of generalized inequalities

- $K = \mathbb{R}_+^n$ gives usual partial ordering on $\mathbb{R}^n$ (componentwise)



$$x \preceq_K y \iff x_i \leq y_i, \forall i$$

- $K = \mathbb{S}_+^n$ = set of **symmetric positive semidefinite matrices**

$$A \preceq B \iff B - A \in \mathbb{S}_+^n$$

- K = **cone of positive polynomials**

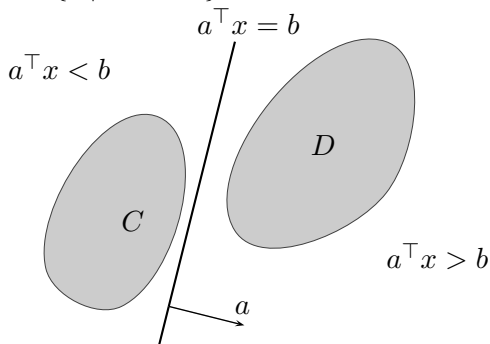
$$p \preceq_K q \iff 0 \leq (q_1 - p_1) + (q_2 - p_2)t + \cdots + (q_n - p_n)t^{n-1}, \forall t$$

Separating hyperplane theorem

If C and D are disjoint convex sets ($C \cap D = \emptyset$),
there exist $a \neq 0, a \in \mathbb{R}^n$ and $b \in \mathbb{R}$ such that:

$$\forall x \in C, a^\top x \leq b \quad \text{and} \quad \forall x \in D, a^\top x \geq b$$

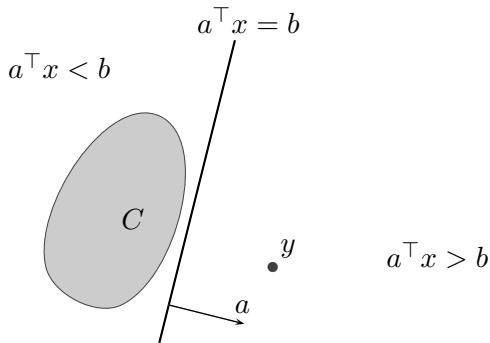
The hyperplane $\{x \mid a^\top x = b\}$ separates C and D .



Strict separation

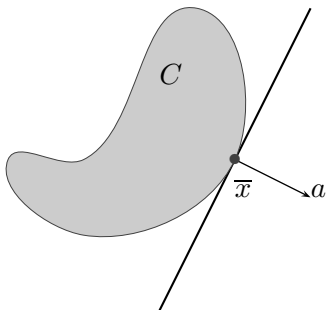
Basic separation (point and set) If C closed and convex and $y \notin C$, there exist $a \neq 0, a \in \mathbb{R}^n$ and $b \in \mathbb{R}$ such that:

$$\forall x \in C, \quad a^\top x \leq b < a^\top y$$



Supporting hyperplane

$C \subset \mathbb{R}^n$, $\bar{x} \in \text{bd } C$ If $a \neq 0$ and $\forall x \in C, a^\top x \leq a^\top \bar{x}$, then $\{x \in \mathbb{R}^n \mid a^\top x = a^\top \bar{x}\}$ is a **supporting hyperplane** of C .



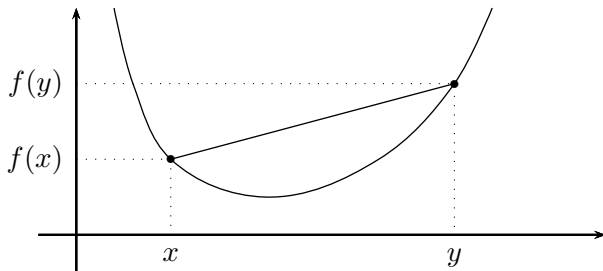
→ If C is convex, then there exist a supporting hyperplane at every boundary point of C .

Convex function

$f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **convex** if $\text{dom } f$ is a convex set and

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

for all $x, y \in \text{dom } f$, $0 \leq \theta \leq 1$.

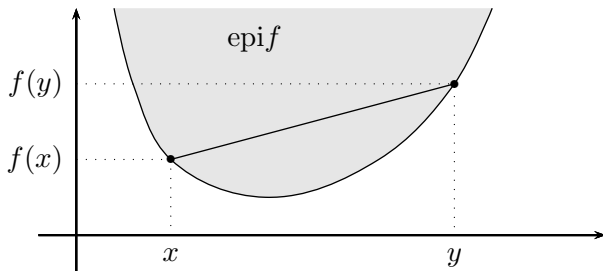


- **strictly convex** when: $f(\theta x + (1 - \theta)y) < \theta f(x) + (1 - \theta)f(y)$
- f is **concave** if $(-f)$ is convex.

Epigraph

The **epigraph** of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is:

$$\text{epi} f := \{(x, t) \in \mathbb{R}^{n+1} \mid x \in \text{dom } f, f(x) \leq t\}$$



- f is convex if and only if its epigraph is convex.
- sublevel set: $C_\alpha := \{x \mid f(x) \leq \alpha\}$
- ▷ C_α is a convex set if f convex

Jensen's inequality

For a convex function f :

- $f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$: called **Jensen's inequality**
- extends to
 - ▶ sums (finite or not): for $\theta_1, \dots, \theta_p \geq 0$, $\theta_1 + \dots + \theta_p = 1$:

$$f(\theta_1 x_1 + \dots + \theta_p x_p) \leq \theta_1 f(x_1) + \dots + \theta_p f(x_p)$$

- ▶ integrals and expected values: if $p(x)$ is a pdf with support $S \subset \text{dom } f$:

$$f\left(\int_S x p(x) \, dx\right) \leq \int_S f(x) p(x) \, dx \quad f(\mathbb{E}\{X\}) \leq \mathbb{E}\{f(X)\}$$

Examples of convex/concave functions

convex

- $\|x\|$
- $\max(x_1, \dots, x_n)$
- $f(x, y) = \frac{x^2}{y}$ with $\text{dom } f = \mathbb{R} \times \mathbb{R}_{++}$
- $\log(e^{x_1} + \dots + e^{x_n})$

concave

- $f(x) = (\prod_{i=1}^n x_i)^{1/n}$
- $f(X) = \log \det X$ with $\text{dom } f = \mathbb{S}_{++}^n$.

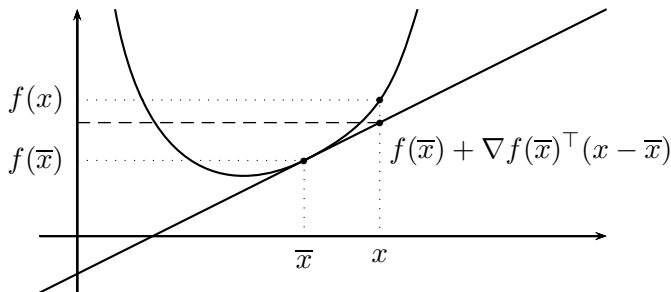
convex and concave

- affine functions: $f(x) = a^\top x + b$

First order conditions

Differentiable f with convex domain is convex if and only if:

$$f(x) \geq f(\bar{x}) + \nabla f(\bar{x})^\top (x - \bar{x}) \quad \forall x, \bar{x} \in \text{dom } f$$



The linear approximation of f is a global underestimator.

Second order conditions

Twice differentiable f with convex domain:

$$f \text{ convex} \Leftrightarrow \nabla^2 f(x) \succeq 0 \quad \forall x \in \text{dom } f$$

If $\nabla^2 f(x) \succ 0 \quad \forall x \in \text{dom } f$, then f strictly convex.

- Ex: $f(x) = \frac{1}{2}x^\top Px + q^\top x + r$ defined on $\mathbb{R}^n$ is:
 - ▶ convex iff $P \succeq 0$ (concave iff $P \preceq 0$),
 - ▶ strictly convex iff $P \succ 0$ (strictly concave iff $P \prec 0$).

Operations that preserve convexity

(1/3)

Nonnegative weighted sums: $f = w_1 f_1 + \dots + w_m f_m$ is convex if $f_1, \dots, f_m$ convex and $w_1, \dots, w_m \geq 0$.

Composition with an affine mapping: $x \mapsto f(Ax + b)$ is convex (resp. concave) if f convex (resp. concave)

Pointwise maximum: $x \mapsto \max\{f_1(x), \dots, f_m(x)\}$ is convex if $f_1, \dots, f_m$ convex (extends to supremum).

Operations that preserve convexity

(2/3)

Composition: let $g : \mathbb{R}^n \rightarrow \mathbb{R}^k$ and $h : \mathbb{R}^k \rightarrow \mathbb{R}$ and $f = h \circ g : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $f(x) = h(g(x))$.

- f is convex if h is convex nondecreasing and g is convex,
- f is convex if h is convex nonincreasing and g is concave,
- f is concave if h is concave nondecreasing and g is concave,
- f is concave if h is concave nonincreasing and g is convex.

(Easy proof in simple real valued differentiable case.)

Operations that preserve convexity

(3/3)

Minimization: if $f(x, y)$ convex in (x, y) , $C \neq \emptyset$, $g(x) = \inf_{y \in C} f(x, y)$ is convex in x provided $g(x) > -\infty$ for some x .

Perspective of a function: perspective function of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is $g : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ defined by

$$g(x, t) = tf(x/t)$$

The perspective preserves convexity.

How to prove convexity?

- ① verify definition, often simplified by restricting to a line:
 - ▷ f is convex if and only if it is convex when restricted to any line that intersects $\text{dom } f$
Ex: prove concavity of $f(X) = \log \det X$ with $\text{dom } f = \mathbb{S}_{++}^n$.
- ② for twice differentiable functions, second-order condition
- ③ show that f is obtained from simple convex functions by operations that preserve convexity.

Optimization problem in standard form

General form, **non convex** (but can be):

$$\begin{cases} \min. & f(x) \quad (x \in \mathcal{D} \subset \mathbb{R}^n) \\ \text{s.t.} & g_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_j(x) = 0, \quad j = 1, \dots, p \end{cases}$$

- $x = (x_1, \dots, x_n)^\top$: optimization variables
- $f : \mathcal{D} \rightarrow \mathbb{R}$: objective or cost function
- $g_i : \mathcal{D} \rightarrow \mathbb{R}, i = 1, \dots, m$: inequality constraint functions
- $h_j : \mathcal{D} \rightarrow \mathbb{R}, j = 1, \dots, p$: equality constraint functions

optimal value: $p^\star := \inf\{f(x) | g_i(x) \leq 0, h_j(x) = 0, x \in \mathcal{D}\}$

- $p^\star = +\infty$: problem unfeasible (no x satisfies the constraints)
- $p^\star = -\infty$: problem unbounded below

Vocabulary, remarks

- Constraints:
 - ▶ **implicit**: $x \in \mathcal{D}$ intersection of all functions domain:
 $\mathcal{D} \subset \text{dom } f$, $\mathcal{D} \subset \text{dom } g_i$ and $\mathcal{D} \subset \text{dom } h_j$
 - ▶ **explicit**: $g_i(x) \leq 0$, $h_j(x) = 0$
 - ▶ **unconstrained** problem: only implicit constraints
- **Feasible point**: any x that satisfies the constraint.
 - ▶ feasibility problem = find a feasible point = special case of general problem with $f(x) = 0$
- optimal point x^* :
 - ▶ x^* **global optimal** if feasible and $p^* = f(x^*) \leq f(x)$ for any feasible x
 - ▶ x_{loc}^* **local optimum** if feasible and $f(x_{\text{loc}}^*) \leq f(x)$ for any x such that $\|x - x_{\text{loc}}^*\| \leq \alpha$ and x feasible.

Convex optimization problem (standard form)

$$\begin{cases} \min. f(x) & (x \in \mathcal{D} = (\cap_{i=1}^m \text{dom } g_i) \cap \text{dom } f) \\ \text{s.t. } g_i(x) \leq 0, & i = 1, \dots, m \\ a_i^\top x = b_i, & i = 1, \dots, p \end{cases}$$

- **objective** f and **constraint** functions $g_1, \dots, g_m$ are convex
- **equality constraints** are **affine**.

often written as:

$$\begin{cases} \min. f(x) & (x \in \mathcal{D}) \\ \text{s.t. } g_i(x) \leq 0, & i = 1, \dots, m \\ Ax = b \end{cases}$$

Remark: can be written **with inequalities only**.

Indeed, for $i = 1, \dots, p$, replace the equalities by the two inequalities $a_i^\top x - b_i \leq 0$ and $-a_i^\top x + b_i \leq 0$

Feasible set of a convex optimization problem

- General convex problem with inequalities only:

$$\begin{cases} \min. f(x) & (x \in \mathcal{D}) \\ \text{s.t. } g_i(x) \leq 0, & i = 1, \dots, m \end{cases}$$

- for all i , the sublevel set $C_i = \{x \in \mathbb{R}^n \mid g_i(x) \leq 0\}$ is convex (follows from convexity of g_i)
- **feasible set** $X := \mathcal{D} \cap \bigcap_{i=1}^m C_i$ is **convex**
- A convex optimization problem minimizes a convex function over a convex set

(take care: some convex sets may be nasty and intractable.
E.g. polynomials positive on a set)

Global / local optimality for a convex optimization problem

Any **locally** optimal point of a **convex** problem is **globally** optimal.

Proof: Let x_{loc}^* be a local optimum. For an $R > 0$,

$$\forall x \text{ feasible, } \|x - x_{\text{loc}}^*\| < R \Rightarrow f(x_{\text{loc}}^*) \leq f(x).$$

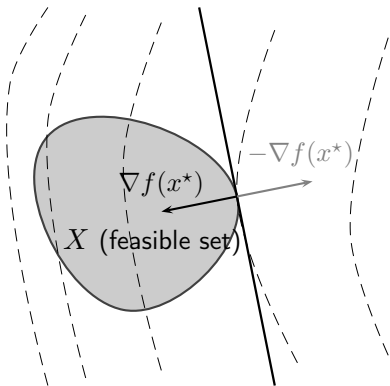
x_{loc}^* not global $\Rightarrow f(\bar{x}) < f(x_{\text{loc}}^*)$ for a feasible $\bar{x}$.

Let $z = (1 - \theta)x_{\text{loc}}^* + \theta\bar{x}$ with $\theta = \frac{R}{2\|\bar{x} - x_{\text{loc}}^*\|} < 1$ and use convexity to get a contradiction:

$$f(x_{\text{loc}}^*) \leq f(z) \leq (1 - \theta)f(x_{\text{loc}}^*) + \theta f(\bar{x}) < f(x_{\text{loc}}^*)$$

Optimality criterion

For convex and differentiable f ($\text{dom } f$ open), x^* is optimal if and only if:
 x^* feasible and for all feasible x , $\nabla f(x^*)^\top (x - x^*) \geq 0$.



Proof: condition is

- sufficient because of convexity:

$$f(x) \geq f(x^*) + \nabla f(x^*)^\top (x - x^*)$$
- necessary: $h(t) = f((1-t)x^* + tx)$
 then:

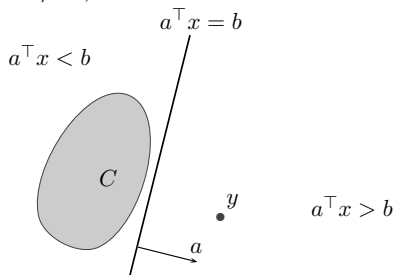
$$h'(0) = \nabla f(x^*)^\top (x - x^*)$$

 $\rightsquigarrow f$ can be decreased if < 0 for a feasible x

- Equivalent condition: $-\nabla f(x^*) \in \mathcal{N}_X(x^*)$ (normal cone)

Strict separation

Basic separation (point and set) If C closed and convex and $y \notin C$, there exist $a \neq 0, a \in \mathbb{R}^n$ and $b \in \mathbb{R}$ such that $\forall x \in C, \quad a^\top x \leq b < a^\top y$.



Proof: Let $\bar{x}$ be a minimizer of $f(x) = \frac{\|x-y\|^2}{2}$ on C (which exists).

Optimality condition $-\nabla f(\bar{x}) \in \mathcal{N}_C(\bar{x})$, yields for all $x \in C$

$$(y - \bar{x})^\top (x - \bar{x}) \leq 0 \quad \text{that is:}$$

$$\underbrace{(y - \bar{x})^\top x}_{=:a} \leq \underbrace{(y - \bar{x})^\top \bar{x}}_{=:b} < (y - \bar{x})^\top y.$$

Optimality criterion

(examples, see the exercises)

Particular cases, with differentiable f ($\text{dom } f$ open):

- unconstrained problem: $\min. f(x)$

$$x^* \text{ optimal} \Leftrightarrow \nabla f(x^*) = 0, \quad x^* \in \text{dom } f$$

- equality constrained problem:
$$\begin{cases} \min. f(x) \\ \text{s.t. } Ax = b \end{cases}$$

$$x^* \text{ optimal} \Leftrightarrow \nabla f(x^*) + A^\top \nu^* = 0, \quad Ax^* = b, \quad x^* \in \text{dom } f$$

- minimization over nonnegative orthant:
$$\begin{cases} \min. f(x) \\ \text{s.t. } x \succeq 0 \end{cases}$$

$$\begin{aligned} x^* \text{ optimal} \Leftrightarrow \quad & x^* \succeq 0, \quad \nabla f(x^*) \succeq 0, \\ & x_i^* [\nabla f(x^*)]_i = 0, i = 1, \dots, n \end{aligned}$$

Part III

Duality and optimality conditions

Lagrangian (inequality constraints only)

$$\begin{cases} \min. f(x) & x \in \mathcal{D} \subset \mathbb{R}^n \\ \text{s.t. } g_i(x) \leq 0, & i = 1, \dots, m \end{cases}$$

with $\mathcal{D} := \cap_{i=1}^m \text{dom } g_i \cap \text{dom } f$.

Lagrangian $\mathcal{L} : \mathcal{D} \times \mathbb{R}^m \rightarrow \mathbb{R}$

$$\mathcal{L}(x, \lambda) := f(x) + \sum_{i=1}^m \lambda_i g_i(x)$$

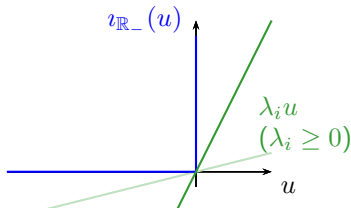
- λ_i are Lagrange multipliers, $\lambda = (\lambda_1, \dots, \lambda_m)^\top$.

Lagrangian: linear approximation interpretation

Equivalent unconstrained form:

$$\min. f(x) + \sum_{i=1}^m v_{\mathbb{R}_-}(g_i(x))$$

Replace indicator functions by “soft” constraint/underestimator:



For $\lambda \succeq 0$:

$$\mathcal{L}(x, \lambda) := f(x) + \sum_{i=1}^m \lambda_i g_i(x) \leq f(x) + \sum_{i=1}^m v_{\mathbb{R}_-}(g_i(x))$$

Lagrange dual function

Dual function

$$\mathcal{L}_D(\lambda) := \inf_{x \in \mathcal{D}} \mathcal{L}(x, \lambda) = \inf_{x \in \mathcal{D}} \left(f(x) + \sum_{i=1}^m \lambda_i g_i(x) \right)$$

- $\mathcal{L}_D$ is **concave** (even if non convex problem), can be $-\infty$
- **Lower bound property**: if $\lambda \succeq 0$, then $\mathcal{L}_D(\lambda) \leq p^*$

Proof: for $\lambda \succeq 0$ and x feasible:

$$\mathcal{L}(x, \lambda) := f(x) + \sum_{i=1}^m \underbrace{\lambda_i}_{\geq 0} \underbrace{g_i(x)}_{\leq 0} \leq f(x)$$

Taking the infimum on the l.h.s yields $\mathcal{L}_D(\lambda) \leq f(x)$ for any feasible x and hence the result.

The dual problem

Lagrange dual problem

$$\begin{cases} \max. \mathcal{L}_D(\lambda) \\ \text{s.t. } \lambda \succeq 0 \end{cases}$$

$$d^* := \sup_{\lambda \succeq 0} \mathcal{L}_D(\lambda)$$

- It is a convex problem
- λ dual feasible if $\lambda \succeq 0, \lambda \in \text{dom } \mathcal{L}_D$

Weak duality: $d^* \leq p^*$ always holds (also for nonconvex problems)
 $p^* - d^*$ is called **duality gap**.

Weak and strong duality

Weak duality (always holds): $d^* \leq p^*$

Strong duality: $d^* = p^*$

- does not hold in general
- holds for convex problems under **constraint qualifications** (see later).

Duality and max-min inequality

Primal with optimal value p^* :
$$\begin{cases} \min. f(x) & (x \in \mathcal{D}) \\ \text{s.t. } g_i(x) \leq 0, & i = 1, \dots, m \end{cases}$$

- Lagrangian: $\mathcal{L}(x, \lambda) := f(x) + \sum_{i=1}^m \lambda_i g_i(x)$
- Primal reads also:

$$p^* = \inf_{x \in \mathcal{D}} \sup_{\lambda \succeq 0} \mathcal{L}(x, \lambda)$$

- Dual problem:

$$d^* = \sup_{\lambda \succeq 0} \inf_{x \in \mathcal{D}} \mathcal{L}(x, \lambda)$$

- We have (max-min inequality):

$$\sup_{\lambda \succeq 0} \inf_{x \in \mathcal{D}} \mathcal{L}(x, \lambda) \leq \inf_{x \in \mathcal{D}} \sup_{\lambda \succeq 0} \mathcal{L}(x, \lambda)$$

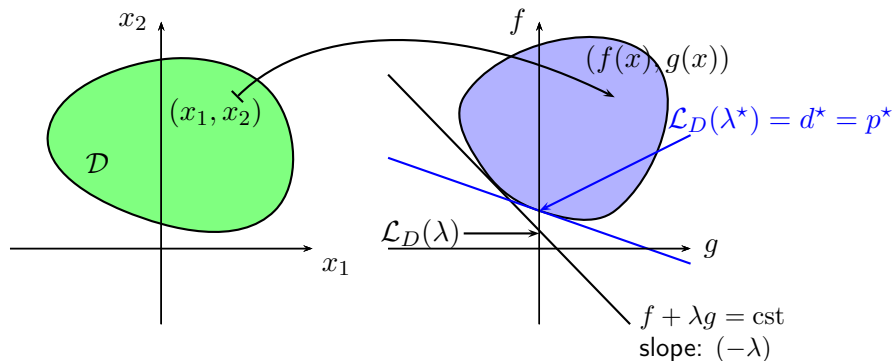
Strong duality when strong max-min/saddle-point property satisfied

Geometric interpretation of duality

Convex case

$$\begin{cases} \min. & f(x) \\ \text{s.t.} & g(x) \leq 0 \end{cases}$$

$$\mathcal{L}_D(\lambda) = \inf_{x \in \mathcal{D}} f(x) + \lambda g(x)$$

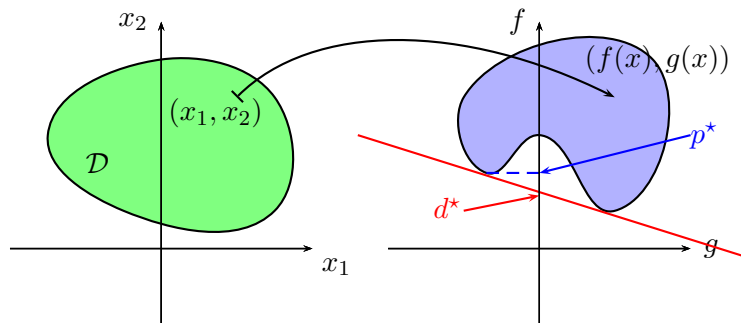


Geometric interpretation of duality

Non-convex case

$$\begin{cases} \min. & f(x) \\ \text{s.t.} & g(x) \leq 0 \end{cases}$$

$$\mathcal{L}_D(\lambda) = \inf_{x \in \mathcal{D}} f(x) + \lambda g(x)$$



Lagrangian (inequality constraints only)

$$\begin{cases} \min. f(x) & x \in \mathcal{D} \subset \mathbb{R}^n \\ \text{s.t. } g_i(x) \leq 0, & i = 1, \dots, m \end{cases}$$

with $\mathcal{D} := \cap_{i=1}^m \text{dom } g_i \cap \text{dom } f$.

Lagrangian $\mathcal{L} : \mathcal{D} \times \mathbb{R}^m \rightarrow \mathbb{R}$

$$\mathcal{L}(x, \lambda) := f(x) + \sum_{i=1}^m \lambda_i g_i(x)$$

- λ_i are Lagrange multipliers, $\lambda = (\lambda_1, \dots, \lambda_m)^\top$.

Lagrangian sufficient conditions

Assume $(x^*, \lambda^*) \in \mathcal{D} \times \mathbb{R}^m$ satisfies:

$$\forall i = 1, \dots, m, \quad g_i(x^*) \leq 0 \quad (\text{primal feasibility})$$

$$\forall i = 1, \dots, m, \quad \lambda_i^* \geq 0 \quad (\text{dual feasibility})$$

$$\forall i = 1, \dots, m, \quad \lambda_i^* g_i(x^*) = 0 \quad (\text{complementary slackness})$$

$$\forall x \text{ feasible}, \quad \mathcal{L}(x^*, \lambda^*) \leq \mathcal{L}(x, \lambda^*) \quad (x^* \text{ minimizes } \mathcal{L}(., \lambda^*))$$

then, x^* is optimal (global minimum).

Proof: For any feasible x :

$$f(x^*) = \mathcal{L}(x^*, \lambda^*) \leq \mathcal{L}(x, \lambda^*) = f(x) + \underbrace{\sum_{i=1}^m \lambda_i^* g_i(x)}_{\leq 0} \leq f(x)$$

- λ^* : Lagrange multiplier vector
- Remark: no convexity!

KKT conditions (Karush-Kuhn-Tucker)

Convex case: sufficient conditions

Assume $(x^*, \lambda^*) \in \text{int } \mathcal{D} \times \mathbb{R}^m$ satisfies:

$$\forall i = 1, \dots, m, \quad g_i(x^*) \leq 0 \quad (\text{primal feasibility})$$

$$\forall i = 1, \dots, m, \quad \lambda_i^* \geq 0 \quad (\text{dual feasibility})$$

$$\forall i = 1, \dots, m, \quad \lambda_i^* g_i(x^*) = 0 \quad (\text{complementary slackness})$$

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(x^*) = 0 \quad (x^* \text{ critical point of the Lagrangian})$$

then, if the problem is **convex**, x^* is optimal.

- λ^* : Lagrange multiplier vector
- Remark: for convex functions $f, g_1, \dots, g_m$, last condition implies $\mathcal{L}(x^*, \lambda^*) \leq \mathcal{L}(x, \lambda^*)$

Necessary optimality conditions (Fritz-John)

$$\begin{cases} \min. f(x) & x \in \mathcal{D} \subset \mathbb{R}^n \\ \text{s.t. } g_i(x) \leq 0, & i = 1, \dots, m \end{cases}$$

- **Active set** at point x : $I(x) = \{i \in \{1, \dots, m\} \mid g_i(x) = 0\}$
- Fritz-John optimality conditions:
If $x_{\text{loc}}^* \in \text{int } \mathcal{D}$ is a **local minimizer**, there exist $\lambda_0, \lambda_1, \dots, \lambda_m \geq 0$ such that:

$$\lambda_0 \nabla f(x_{\text{loc}}^*) + \sum_{i \in I(x_{\text{loc}}^*)} \lambda_i \nabla g_i(x_{\text{loc}}^*) = 0$$

- For $i \notin I(x_{\text{loc}}^*)$, complementary slackness yields $\lambda_i = 0 \rightsquigarrow$ terms don't appear above.
- To rule out the case $\lambda_0 = 0$, **constraint qualification** at x_{loc}^* (required for KKT to be necessary conditions)

Local constraint qualifications

Constraint qualifications **at a point** x :

- MFCQ (Mangasarian-Fromovitz constraint qualification):
there is a direction d satisfying $\nabla g_i(x)^\top d < 0$ for all $i \in I(x)$
- LICQ (linear independence constraint qualification):
 $\{\nabla g_i(x)\}_{i \in I(x)}$ are linearly independent

Property: LICQ $\Rightarrow$ MFCQ

Global constraint qualification (Slater)

- **Slater** constraint qualification for convex problem with constraints $g_i(x) \leq 0, \quad i = 1, \dots, m$
 - there exists $\hat{x} \in \text{relint } \mathcal{D}$ with $g_i(\hat{x}) < 0, \quad i = 1, \dots, m$
 - Refinement: **affine inequalities need not be strict**. For constraints
$$\begin{cases} g_i(x) \leq 0, & i = 1, \dots, m \\ Ax \leq b, & Cx = d \end{cases}$$
 - there exists $\hat{x} \in \text{relint } \mathcal{D}$ with $g_i(\hat{x}) < 0, \quad i = 1, \dots, m$ and $Ax \leq b, Cx = d$
- + For a convex problem: Slater $\Rightarrow$ MFCQ at any feasible point.
- + Slater $\approx$ there exist a strictly feasible point
- + Slater $\Rightarrow$ strong duality and dual value attained when $d^* > -\infty$

KKT necessary optimality conditions

Suppose x_{loc}^* is a **local minimizer** of

$$\inf\{f(x) \mid x \in \mathcal{D}, g_i(x) \leq 0, i = 1, \dots, m\}$$

If MFCQ holds at x_{loc}^* , there is a Lagrange multiplier vector λ^* for x_{loc}^* :

$$\forall i = 1, \dots, m, \quad g_i(x_{\text{loc}}^*) \leq 0 \quad (\text{primal feasibility})$$

$$\forall i = 1, \dots, m, \quad \lambda_i^* \geq 0 \quad (\text{dual feasibility})$$

$$\forall i = 1, \dots, m, \quad \lambda_i^* g_i(x_{\text{loc}}^*) = 0 \quad (\text{complementary slackness})$$

$$\nabla f(x_{\text{loc}}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(x_{\text{loc}}^*) = 0 \quad (x_{\text{loc}}^* \text{ critical point of the Lagrangian})$$

Remarks:

- No convexity here, but local minimizer considered.
- For convex problems, above conditions are necessary and sufficient for global optimality.

Perturbation and sensitivity analysis (1/2)

- Unperturbed optimization problem and dual

$$p^* : \begin{cases} \min. f(x) \\ \text{s.t. } g_i(x) \leq 0, \quad 1 \leq i \leq m \end{cases} \quad \begin{cases} \max. \mathcal{L}_D(\lambda) \\ \text{s.t. } \lambda \succeq 0 \end{cases}$$

- Perturbed problem and dual

$$p^*(u) : \begin{cases} \min. f(x) \\ \text{s.t. } g_i(x) \leq u_i, \quad 1 \leq i \leq m \end{cases} \quad \begin{cases} \max. \mathcal{L}_D(\lambda) - u^\top \lambda \\ \text{s.t. } \lambda \succeq 0 \end{cases}$$

Optimal value $p^*(u)$ as a function of parameters u
(for the original problem $p^* = p^*(0)$)

Perturbation and sensitivity analysis (2/2)

Assume for problem, strong duality and λ^* dual optimal.

- Global sensitivity:

$$\begin{aligned} p^*(u) &\geq \mathcal{L}_D(\lambda^*) - u^\top \lambda^* \text{ (weak duality pert. prob.)} \\ &\geq p^*(0) - u^\top \lambda^* \text{ (strong duality)} \end{aligned}$$

- Local sensitivity: if $p^*(u)$ differentiable at 0:

$$\lambda_i^* = -\frac{\partial p^*(0)}{\partial u_i}$$

Proof: take $u = te_i$ where e_i is i^{th} canonical basis vector and get $\frac{p^*(te_i) - p^*(0)}{t} \geq -\lambda_i^*$ for $t > 0$ or $\leq -\lambda_i^*$ for $t < 0$.

- Interpretation: ...

Lagrangian and dual function

$$\begin{cases} \min. f(x) & (x \in \mathcal{D} \subset \mathbb{R}^n) \\ \text{s.t. } g_i(x) \leq 0, & i = 1, \dots, m \\ \quad \quad \quad h_j(x) = 0, & j = 1, \dots, p \end{cases}$$

Lagrangian $\mathcal{L} : \mathcal{D} \times \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$

(λ_i, ν_j are Lagrange multipliers)

$$\mathcal{L}(x, \lambda, \nu) := f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x)$$

Dual function $\mathcal{L}_D(\lambda, \nu) := \inf_{x \in \mathcal{D}} \mathcal{L}(x, \lambda, \nu)$

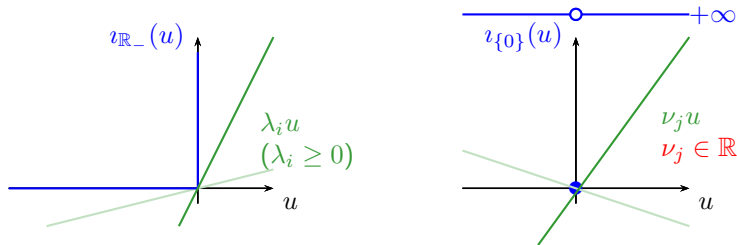
- $\mathcal{L}_D$ is **concave** (even if non convex problem), can be $-\infty$
- **Lower bound property**: if $\lambda \succeq 0$, $\nu \in \mathbb{R}^p$, then $\mathcal{L}_D(\lambda, \nu) \leq p^*$

Lagrangian: linear approximation interpretation

Equivalent unconstrained form:

$$\min. f(x) + \sum_{i=1}^m v_{\mathbb{R}_-}(g_i(x)) + \sum_{j=1}^p v_{\{0\}}(h_j(x))$$

Replace indicator functions by “soft” constraint/underestimator:



For $\lambda \succeq 0$ and $\nu \in \mathbb{R}^p$,

$$\mathcal{L}(x, \lambda, \nu) \leq f(x) + \sum_{i=1}^m v_{\mathbb{R}_-}(g_i(x)) + \sum_{j=1}^p v_{\{0\}}(h_j(x)).$$

Lagrange dual function

Lagrange dual problem

$$d^* := \sup_{\lambda \succeq 0, \nu \in \mathbb{R}^p} \mathcal{L}_D(\lambda, \nu) = \begin{cases} \max. \mathcal{L}_D(\lambda, \nu) \\ \text{s.t. } \lambda \succeq 0 \end{cases}$$

- It is a convex problem.
- λ, ν are dual feasible if $\lambda \succeq 0, \nu \in \mathbb{R}^p, (\lambda, \nu) \in \text{dom } \mathcal{L}_D$

Weak duality (always holds): $d^* \leq p^*$

$p^* - d^*$ is called **duality gap**.

Strong duality: $d^* = p^*$

- does not hold in general.
- holds for convex problems under **constraint qualifications**.

Duality and max-min inequality

Primal with optimal value p^* :
$$\begin{cases} \min. f(x) & (x \in \mathcal{D}) \\ \text{s.t. } g_i(x) \leq 0, & i = 1, \dots, m \\ h_j(x) = 0, & j = 1, \dots, p \end{cases}$$

- Lagrangian: $\mathcal{L}(x, \lambda, \nu) := f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x)$
- Primal reads also:

$$p^* = \inf_{x \in \mathcal{D}} \sup_{\nu \in \mathbb{R}^p, \lambda \succeq 0} \mathcal{L}(x, \lambda, \nu)$$

- Dual problem:

$$d^* = \sup_{\nu \in \mathbb{R}^p, \lambda \succeq 0} \inf_{x \in \mathcal{D}} \mathcal{L}(x, \lambda, \nu)$$

- We have (max-min inequality):

$$d^* = \sup_{\nu \in \mathbb{R}^p, \lambda \succeq 0} \inf_{x \in \mathcal{D}} \mathcal{L}(x, \lambda, \nu) \leq \inf_{x \in \mathcal{D}} \sup_{\nu \in \mathbb{R}^p, \lambda \succeq 0} \mathcal{L}(x, \lambda, \nu) = p^*$$

Strong duality when strong max-min/saddle-point property satisfied.

KKT optimality conditions

$$g_i(x^*) \leq 0, \quad i = 1, \dots, m \quad (\text{primal feasibility})$$

$$h_j(x^*) = 0, \quad j = 1, \dots, p$$

$$\lambda_i^* \geq 0, \quad i = 1, \dots, m \quad (\text{dual feasibility})$$

$$(\nu_j^* \in \mathbb{R}, \quad j = 1, \dots, p)$$

$$\lambda_i^* g_i(x^*) = 0, \quad i = 1, \dots, m \quad (\text{complementary slackness})$$

$$\mathcal{L}(x^*, \lambda^*, \nu^*) \leq \mathcal{L}(x, \lambda^*, \nu^*), \quad \forall x \text{ feasible} \quad (x^* \text{ minimizes } \mathcal{L}(., \lambda^*, \nu^*))$$

KKT optimality conditions

$$g_i(x^*) \leq 0, \quad i = 1, \dots, m \quad (\text{primal feasibility})$$

$$h_j(x^*) = 0, \quad j = 1, \dots, p$$

$$\lambda_i^* \geq 0, \quad i = 1, \dots, m \quad (\text{dual feasibility})$$

$$(\nu_j^* \in \mathbb{R}, \quad j = 1, \dots, p)$$

$$\lambda_i^* g_i(x^*) = 0, \quad i = 1, \dots, m \quad (\text{complementary slackness})$$

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(x^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(x^*) = 0 \quad (x^* \text{ critical point of the Lagrangian})$$

- Remark: for convex problem, last condition implies $\mathcal{L}(x^*, \lambda^*, \nu^*) \leq \mathcal{L}(x, \lambda^*, \nu^*)$ for feasible x .

Least-norm solution of linear equation (example)

Lagrange dual

$$\min. \|x\|_2^2 \quad \text{s.t. } Ax = b$$

- Lagrangian:

$$\mathcal{L}(x, \nu) = x^\top x + \nu^\top (Ax - b)$$

- Dual function: (minimum of $\mathcal{L}$ w.r.t. x when $\nabla_x \mathcal{L}(x, \nu) = 0$)

$$\begin{aligned}\mathcal{L}_D(\nu) &= \mathcal{L}\left(-\frac{1}{2}A^\top \nu, \nu\right) \\ &= -\frac{1}{4}\nu^\top AA^\top \nu - b^\top \nu \leq \inf\{\|x\|_2^2 \mid Ax = b\}\end{aligned}$$

- Primal and dual problems:

$$p^* : \begin{cases} \min. x^\top x \\ \text{s.t. } Ax = b \end{cases} \qquad d^* : \max. -\frac{1}{4}\nu^\top AA^\top \nu - b^\top \nu$$

Least-norm solution of linear equation (example)

KKT conditions and solution

$$\min. \|x\|_2^2 \quad \text{s.t. } Ax = b$$

- Lagrangian: $\mathcal{L}(x, \nu) = x^\top x + \nu^\top (Ax - b)$
- Dual function: $\mathcal{L}_D(\nu) = -\frac{1}{4}\nu^\top AA^\top \nu - b^\top \nu$
- KKT conditions:

$$\begin{cases} Ax^\star = b \\ 2x^\star + A^\top \nu^\star = 0 \end{cases}$$

- Solution (when $AA^\top$ invertible):

$$\begin{cases} x^\star = A^\top (AA^\top)^{-1} b \\ \nu^\star = -2(AA^\top)^{-1} b \end{cases}$$

LP (standard form) (example)

Lagrange dual

$$\min. c^\top x \quad \text{s.t. } Ax = b, x \succeq 0$$

- Lagrangian:

$$\begin{aligned}\mathcal{L}(x, \lambda, \nu) &= c^\top x - \lambda^\top x + \nu^\top (Ax - b) \\ &= -b^\top \nu + (c + A^\top \nu - \lambda)^\top x\end{aligned}$$

- Dual function:

$$\mathcal{L}_D(\lambda, \nu) = \begin{cases} -b^\top \nu & \text{if } A^\top \nu - \lambda + c = 0 \\ -\infty & \text{otherwise.} \end{cases}$$

- Primal and dual problems:

$$p^* : \begin{cases} \min. c^\top x \\ \text{s.t. } Ax = b \\ x \succeq 0 \end{cases} \quad d^* : \begin{cases} \max. -b^\top \nu \\ \text{s.t. } A^\top \nu + c \succeq 0 \end{cases}$$

LP (standard form) (example)

KKT conditions

$$\min. c^\top x \quad \text{s.t. } Ax = b, x \succeq 0$$

- Lagrangian:

$$\begin{aligned}\mathcal{L}(x, \lambda, \nu) &= c^\top x - \lambda^\top x + \nu^\top (Ax - b) \\ &= -b^\top \nu + (c + A^\top \nu - \lambda)^\top x\end{aligned}$$

- KKT conditions:

$$\begin{cases} Ax^* = b, & x^* \succeq 0 \\ \lambda^* \succeq 0 \\ \lambda_i^* x_i^* = 0, & i = 1, \dots, n \\ A^\top \nu^* + c - \lambda^* = 0 \end{cases}$$

Equality constr. convex quad. minimization (example)

KKT conditions

$$\begin{cases} \min. \frac{1}{2}x^\top Px + q^\top x + r \\ \text{s.t. } Ax = b \end{cases} \quad \text{with } P \in \mathbb{S}_+^n.$$

- Lagrangian: $\mathcal{L}(x, \nu) = \frac{1}{2}x^\top Px + q^\top x + r + \nu^\top (Ax - b)$
- KKT conditions:

$$Ax^\star = b, \quad Px^\star + q + A^\top \nu^\star = 0$$

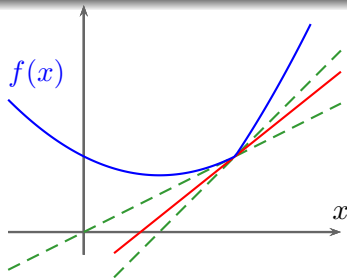
can be written as:

$$\begin{bmatrix} P & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} x^\star \\ \nu^\star \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}$$

Subgradient

A **subgradient** of f at $\bar{x}$ is any vector ϕ such that:

$$f(\bar{x}) + \phi^\top(x - \bar{x}) \leq f(x) \text{ for all } x$$



- $x \mapsto f(\bar{x}) + \phi^\top(x - \bar{x})$ is a *linear underestimator* of f .
- if f convex and differentiable, $\nabla f(\bar{x})$ is (unique) subgradient.

Subdifferential

The **subdifferential** of f at $\bar{x}$ is the set of all subgradients, denoted by:

$$\partial f(\bar{x}) = \{\phi \in \mathbb{R}^n \mid f(\bar{x}) + \phi^\top (x - \bar{x}) \leq f(x) \text{ for all } x\}$$

- $\partial f(\bar{x})$ is a closed convex set (always).
- ∂f is a multi-function / set-valued map. $\partial f : \mathbb{R}^n \rightarrow 2^{\mathbb{R}^n}$
- $\text{dom } \partial f = \{x \in \mathbb{R}^n \mid \partial f(x) \neq \emptyset\}$.
- If f convex and differentiable at $\bar{x} \in \text{int dom } f$, then $\partial f(\bar{x}) = \{\nabla f(\bar{x})\}$.

Subdifferential

Examples

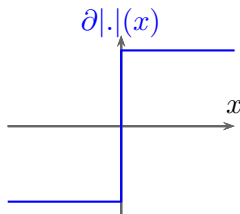
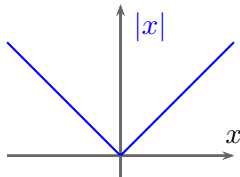
The **subdifferential** of f at $\bar{x}$ is the set of all subgradients:

$$\partial f(\bar{x}) = \{\phi \in \mathbb{R}^n \mid f(\bar{x}) + \phi^\top (x - \bar{x}) \leq f(x) \text{ for all } x\}$$

- If f convex and differentiable at $\bar{x} \in \text{int dom } f$, then

$$\partial f(\bar{x}) = \{\nabla f(\bar{x})\}.$$

- Absolute value: $\partial | \cdot | (x) = \begin{cases} \{-1\} & x < 0, \\ \{1\} & x > 0, \\ [-1, 1] & x = 0 \end{cases}$



- Indicator function: $\partial \iota_C(\bar{x}) = \mathcal{N}_C(\bar{x})$ (normal cone operator)

Fermat's rule

Characterization of global minimizer:

$$x^* \text{ global minimizer of } f \Leftrightarrow 0 \in \partial f(x^*)$$

Proof: Use definition of subdifferential:

$$0 \in \partial f(x^*) \Leftrightarrow f(x^*) + \langle 0, x - x^* \rangle \leq f(x) \text{ for all } x$$

Remark:

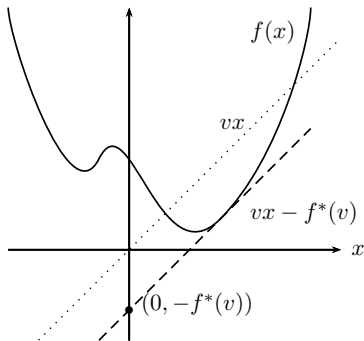
- holds also for nonconvex f .

The Fenchel conjugate function (definition)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ (with $f(x) = \infty$ for $x \notin \text{dom } f$).

Fenchel conjugate (definition)

$$f^*(v) = \sup_{x \in \mathbb{R}^n} (v^\top x - f(x))$$



For f differentiable, $\sup_x (vx - f(x))$ attained at $\bar{x}$ such that $v = f'(\bar{x})$.

Dashed line equation:
 $x \mapsto v(x - \bar{x}) + f(\bar{x}) = vx - f^*(v)$.

The Fenchel conjugate function (first properties)

$$f^*(v) = \sup_{x \in \mathbb{R}^n} (v^\top x - f(x))$$

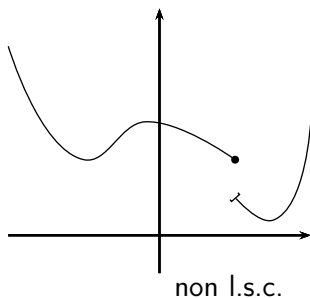
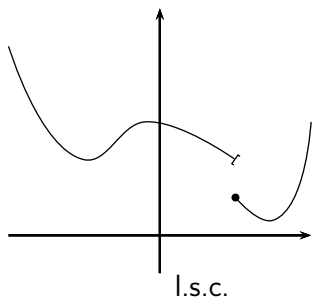
- f^* is convex
(because sup of affine functions. True for non convex f also.)
- $f \geq g$ implies $f^* \leq g^*$
- If $\text{dom } f \neq \emptyset$, f^* never takes the value $-\infty$
- f^* is l.s.c. (lower semi-continuous)
(because epigraph closed as intersection of closed half-spaces)

Lower semi-continuous function (l.s.c.)

f is l.s.c. if and only if at any point x :

$$x_n \xrightarrow{n \rightarrow \infty} x \Rightarrow f(x) \leq \lim_{n \rightarrow \infty} f(x_n)$$

f is l.s.c. $\Leftrightarrow$ epigraph $\{(x, t) \in \mathbb{R}^n \times \mathbb{R} \mid f(x) \leq t\}$ is a closed set



Remark: sometimes, l.s.c. function = closed function

Fenchel conjugate function

Examples

- $f(x) = ax + b$

$$f^*(v) = \begin{cases} -b & \text{if } v = a, \\ +\infty & \text{otherwise.} \end{cases}$$

- $f(x) = e^x$

$$f^*(v) = \begin{cases} v \log v - v & \text{if } v \geq 0, \\ +\infty & \text{otherwise.} \end{cases}$$

- $f(x) = -\log x$

$$f^*(v) = \begin{cases} -\log(-v) - 1 & \text{if } v < 0, \\ +\infty & \text{otherwise.} \end{cases}$$

Fenchel conjugate function

Examples (continued)

- $f(x) = \frac{1}{2}x^\top Qx$ with $Q \succ 0$

$$f^*(v) = \frac{1}{2}v^\top Q^{-1}v.$$

- $f(x) = \frac{1}{2}\|x\|_2^2$

$$f^*(v) = \frac{1}{2}\|v\|_2^2.$$

- $f(x) = \|x\|$

$$f^*(v) = \begin{cases} 0 & \text{if } \|v\|_* \leq 1, \\ +\infty & \text{otherwise.} \end{cases}$$

Dual norm

Let $\|\cdot\|$ be a norm on $\mathbf{E}$.

Associated **dual norm** $\|\cdot\|_*$

$$\|z\|_* := \sup_{\|x\| \leq 1} \langle z, x \rangle$$

- $\langle z, x \rangle \leq \|z\|_* \|x\|$
- Dual norm of $\|\cdot\|_2$ is itself.
- $\|\cdot\|_\infty$ and $\|\cdot\|_1$ are dual norms of each other.
- Dual of ℓ_p -norm is ℓ_q norm with $\frac{1}{p} + \frac{1}{q} = 1$.
- $\|\cdot\|_{**} = \|\cdot\|$ (need not hold in infinite dimensional spaces)

Fenchel-Young inequality

- For any x and v in $\mathbb{R}^n$: $f(x) + f^*(v) \geq v^\top x$
- Equality case:

$$f(x) + f^*(v) = v^\top x \Leftrightarrow v \in \partial f(x)$$

Example: $f(x) = \frac{1}{2}\|x\|^2$ $f^*(v) = \frac{1}{2}\|v\|^2$

$$\rightarrow \frac{1}{2}\|x\|^2 + \frac{1}{2}\|v\|^2 \geq v^\top x$$

Fenchel biconjugate

$$f^{**} = (f^*)^*$$

$$f^{**}(x) = \sup_v (v^\top x - f^*(v))$$

- f^{**} is convex l.s.c.
(from properties of a conjugate)
- f^{**} is a minorant of f
(from Fenchel-Young ineq. $f(x) \geq v^\top x - f^*(v)$ and f^{**} is $\sup_v$ of r.h.s.)

Theorem (Fenchel-Moreau)

$$f = f^{**} \quad \Leftrightarrow \quad f \text{ l.s.c. convex proper}$$

(For proper l.s.c. convex functions, the conjugacy operation induces a bijection.)

- $f^{**} =$ largest l.s.c. convex function bounded above by f
- f l.s.c. convex $\Leftrightarrow f(x) = \sup\{\alpha(x) \mid \alpha \text{ an affine minorant of } f\}$

Subgradient and Fenchel conjugate

Fenchel-Young: $\forall x, v \quad f(x) + f^*(v) \geq v^\top x$

For f **convex, l.s.c., proper**, equality case:

$$\begin{aligned} f(x) + f^*(v) = v^\top x &\Leftrightarrow v \in \partial f(x) \\ &\Leftrightarrow x \in \partial f^*(v) \end{aligned}$$

$\rightarrow \quad \partial f^* = (\partial f)^{-1} \quad \text{Differentiable case: } \nabla f^* = (\nabla f)^{-1}$

Example: With $Q \succ 0$:

$$\begin{aligned} f(x) &= \frac{1}{2} x^\top Q x & f^*(v) &= \frac{1}{2} v^\top Q^{-1} v \\ \nabla f(x) &= Q x & \nabla f^*(v) &= Q^{-1} v \end{aligned}$$

And indeed: $v = \nabla f(x) \Leftrightarrow x = \nabla f^*(v)$

Fenchel duality

Let $f : \mathbb{R}^n \rightarrow]-\infty, +\infty]$, $g : \mathbb{R}^m \rightarrow]-\infty, +\infty]$ and $A \in \mathbb{R}^{m \times n}$.

$$p^* := \inf_{x \in \mathbb{R}^n} \{f(x) + g(Ax)\} \quad (\text{primal value})$$

$$d^* := \sup_{v \in \mathbb{R}^m} \{-f^*(A^\top v) - g^*(-v)\} \quad (\text{dual value})$$

We have:

- Weak duality: $d^* \leq p^*$ (proof: Fenchel-Young inequality)
- Strong duality: if f and g are **convex**, under qualification constraints^a: $p^* = d^*$
and the supremum in the dual problem is attained if finite.

^a $0 \in \text{core}(\text{dom } g - A \text{ dom } f)$ or stronger condition $A \text{ dom } f \cap \text{cont } g \neq \emptyset$

Fenchel and Lagrange duality

- Primal problem: $\min_{x \in \mathbb{R}^n} f(x) + g(Ax)$
- Equivalent constrained problem:

$$\min_{(x,y) \in \mathbb{R}^n \times \mathbb{R}^m} f(x) + g(y) \quad \text{s.t. } y = Ax$$

- Lagrangian $\mathcal{L}(x, y, \nu) = f(x) + g(y) + \nu^\top (y - Ax)$ and dual function:

$$\begin{aligned} \mathcal{L}_D(\nu) &= - \sup_{x \in \mathbb{R}^n} \{x^\top A^\top \nu - f(x)\} - \sup_{y \in \mathbb{R}^m} \{(-\nu)^\top y - g(y)\} \\ &= -f^*(A^\top \nu) - g^*(-\nu) \rightarrow \text{maximizing is Fenchel dual} \end{aligned}$$

- KKT conditions:

$$y^* = Ax^* \quad \nabla f(x^*) - A^\top \nu^* = 0 \quad \nabla g(y^*) + \nu^* = 0 \quad \text{or} \\ \text{equivalently}$$

$$A^\top \nu^* = \nabla f(x^*) \quad -\nu^* = \nabla g(Ax^*)$$

KKT conditions and strong duality

$$p^* = \inf_x f(x) + g(Ax) \quad d^* = \sup_v -f^*(A^\top v) - g^*(-v)$$

Fenchel-Young ineq.:

$$f(x) + f^*(A^\top v) \geq x^\top A^\top v$$

$$g(Ax) + g^*(-v) \geq -x^\top A^\top v$$

$\Rightarrow$

$$f(x) + g(Ax) \geq -f^*(A^\top v) - g^*(-v)$$

Hence **weak** duality:

$$p^* \geq d^*$$

Equality case yield **strong** duality and **KKT** conditions:

$$A^\top v^* \in \partial f(x^*) \quad -v^* \in \partial g(Ax^*)$$

Part IV

Algorithms

Unconstrained minimization

With $f : \mathbb{R}^n \rightarrow \mathbb{R}$ convex, twice differentiable, find solution to:

$$p^* : \quad \min_x f(x)$$

- Optimality condition: $\nabla f(x^*) = 0$
- Produce a **sequence of points** $x^{(k)} \in \text{dom } f$ such that:

$$f(x^{(k)}) \rightarrow p^*$$

- Starting point $x^{(0)}$ required, such that:
 - $x^{(0)} \in \text{dom } f$
 - sublevel set $\{x \mid f(x) \leq f(x^{(0)})\}$ is closed

Descent methods

Starting from $x^{(0)}$ repeat for $k = 0, 1, 2, \dots$:

$$x^{(k+1)} = x^{(k)} + td^{(k)} \quad \text{with } f(x^{(k+1)}) < f(x^{(k)})$$

- $t \geq 0$ is the **step size** or step length
- $d^{(k)}$ is the **search direction** or step. It depends on $x^{(k)}$ and must satisfy:

$$\nabla f(x^{(k)})^\top d^{(k)} < 0$$

(because $f(x^{(k)}) + \nabla f(x^{(k)})^\top (td^{(k)}) \leq f(x^{(k+1)})$ from convexity)

- Simplified notation (sometimes):
current point: x , search direction: d
next point: $x^+ = x + td$ with: $f(x^+) < f(x)$

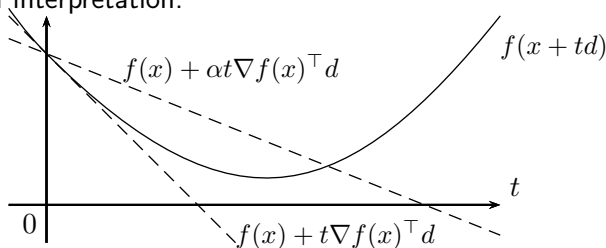
Step size and line search

- **Constant step size** $t > 0$ chosen and fixed.
- **Exact line search** $t = \operatorname{argmin}_{t \geq 0} f(x + td)$
- **Backtracking** (with parameters $\alpha \in]0, 1/2[, \beta \in]0, 1[$) starting at $t = 1$, repeat $t := \beta t$ until:

$$f(x + td) < f(x) + \alpha t \nabla f(x)^\top d$$

(also known as Armijo's rule)

graphical interpretation:



Unconstrained descent method

given starting point $x^{(0)} \in \text{dom } f$, tolerance $\epsilon > 0$,

repeat:

- ❶ Compute search direction $d^{(k)}$
 - ❷ Stopping criterion: **quit** if it is smaller than ϵ .
 - ❸ Choose step size t (backtracking, line search, constant, ...)
 - ❹ Update: $x^{(k+1)} = x^{(k)} + td^{(k)}$
-

Possible search directions for a descent method:

- gradient: $d_{\text{grad}}^{(k)} = -\nabla f(x^{(k)})$
- (normalized) steepest descent: $d_{\text{nsd}}^{(k)} = \operatorname{argmin}_v \{ \nabla f(x^{(k)})^\top v \mid \|v\| \leq 1 \}$
- Newton: $d_{\text{nt}}^{(k)} = -\nabla^2 f(x^{(k)})^{-1} \nabla f(x^{(k)})$

Gradient descent

Gradient descent direction (at point x):

$$d_{\text{grad}} = -\nabla f(x)$$

Stopping condition: usually $\|\nabla f(x^{(k)})\|_2 < \epsilon$.

given starting point $x^{(0)} \in \text{dom } f$, tolerance $\epsilon > 0$,

repeat:

- ❶ Compute gradient $\nabla f(x^{(k)})$.
 - ❷ Stopping criterion: **quit** if $\|\nabla f(x^{(k)})\| < \epsilon$.
 - ❸ Choose step size t (backtracking, line search, constant, ...)
 - ❹ Update: $x^{(k+1)} = x^{(k)} - t\nabla f(x^{(k)})$
-

Convergence of gradient descent

If f is **gradient Lipschitz** (constant $L > 0$):

- fixed step t : convergence if $t \in]0, \frac{2}{L}[$ at rate $\mathcal{O}(1/k)$

If $\mu \mathbf{Id} \preceq \nabla^2 f(x) \preceq L \mathbf{Id}$ (e.g. when **strong convexity**), better results:

- for a constant $c \in]0, 1[$ (which depends on $x^{(0)}$ and f):

$$f(x^{(k)}) - p^\star \leq c^k (f(x^{(0)}) - p^\star)$$

→ **linear convergence**: $f(x^{(k)}) - p^\star \leq \epsilon$ after $\frac{\log((f(x^{(0)}) - p^\star)/\epsilon)}{\log 1/c}$ iterations.

- With exact line search: $c = 1 - \frac{\mu}{L}$

→ Constant c depends on Hessian conditioning and possibly close to 1.

Convergence of gradient descent

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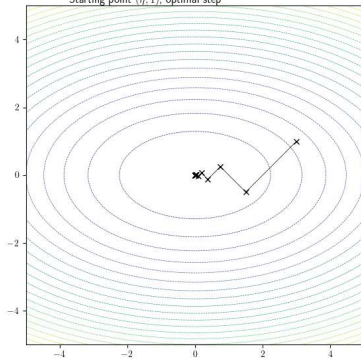
→ Constant c depends on Hessian conditioning and possibly close to 1.

Gradient method:

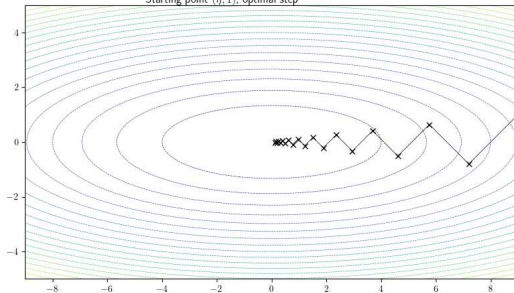
- + simple, cheap iterations
- very slow if problem not well conditioned

Gradient with optimal step

Gradient descent of $f(x, y) = \frac{1}{2}(x^2 + \eta y^2)$ with $\eta = 3.0$
Starting point $(\eta, 1)$, optimal step

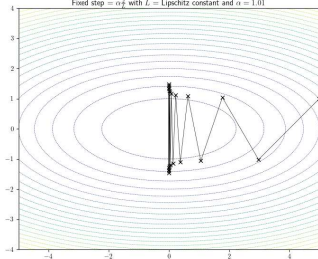


Gradient descent of $f(x, y) = \frac{1}{2}(x^2 + \eta y^2)$ with $\eta = 9.0$
Starting point $(\eta, 1)$, optimal step

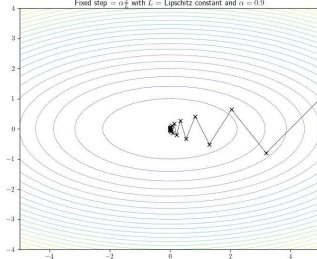


Gradient with fixed step

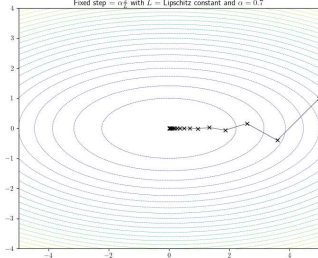
Gradient descent of $f(x, y) = \frac{1}{2}(x^2 + \eta y^2)$ with $\eta = 5.0$
Fixed step = $\alpha \frac{1}{L}$ with $L = \text{Lipschitz constant}$ and $\alpha = 1.01$



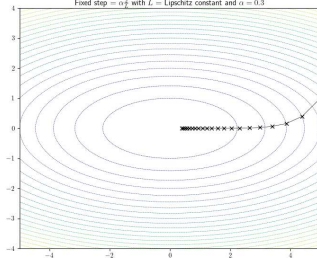
Gradient descent of $f(x, y) = \frac{1}{2}(x^2 + \eta y^2)$ with $\eta = 5.0$
Fixed step = $\alpha \frac{1}{L}$ with $L = \text{Lipschitz constant}$ and $\alpha = 0.9$



Gradient descent of $f(x, y) = \frac{1}{2}(x^2 + \eta y^2)$ with $\eta = 5.0$
Fixed step = $\alpha \frac{1}{L}$ with $L = \text{Lipschitz constant}$ and $\alpha = 0.7$



Gradient descent of $f(x, y) = \frac{1}{2}(x^2 + \eta y^2)$ with $\eta = 5.0$
Fixed step = $\alpha \frac{1}{L}$ with $L = \text{Lipschitz constant}$ and $\alpha = 0.3$



Steepest descent

Normalized direction (at x for given $\|\cdot\|$)

$$d_{\text{nsd}} = \operatorname{argmin}\{\nabla f(x)^\top v \mid \|v\| \leq 1\}$$

Unnormalized direction: $d_{\text{sd}} = \|\nabla f(x)\|_* d_{\text{nsd}}$

- For Euclidian norm, $d_{\text{sd}} = d_{\text{grad}}$.
- For the norm $\|z\|_P = (z^\top P z)^{1/2}$ with $P \in \mathbb{S}_+^n$, $d_{\text{sd}} = -P^{-1} \nabla f(x)$.
- For ℓ_1 norm, $d_{\text{sd}} = -\frac{\partial f(x)}{\partial x_i} e_i$ where e_i is i -th standard basis vector and i such that $\|\nabla f(x)\|_\infty = |[\nabla f(x)]_i|$.

Change of coordinates in gradient descent

Original problem $\min_{x \in \mathbb{R}^n} f(x)$ iterations: $x^{(k+1)} = x^{(k)} - \gamma \nabla f(x^{(k)})$

- $x = Qy$ and $g(y) = f(Qy)$ ($Q \in \mathbb{R}^{n \times n}$ invertible)
- Iterations for equivalent problem $\min_{y \in \mathbb{R}^n} g(y)$:

$$y^{(k+1)} = y^{(k)} - \gamma \underbrace{\nabla g(y^{(k)})}_{Q^\top \nabla f(Qy^{(k)})}$$

that is:

$$\begin{aligned} Qy^{(k+1)} &= Qy^{(k)} - \gamma QQ^\top \nabla f(Qy^{(k)}) \\ x^{(k+1)} &= x^{(k)} - \gamma QQ^\top \nabla f(x^{(k)}) \end{aligned}$$

→ steepest descent in norm given by $(QQ^\top)^{-1}$

Newton step (unconstrained)

Newton method: general descent method with search direction

$$d_{\text{nt}} = -\nabla^2 f(x)^{-1} \nabla f(x).$$

- $x + d_{\text{nt}}$ minimizes second order approximation

$$\hat{f}_2(x + v) = f(x) + \nabla f(x)^\top v + \frac{1}{2} v^\top \nabla^2 f(x) v$$

- $x + d_{\text{nt}}$ solves linearized optimality condition

$$\begin{aligned} \nabla f(x + v) &\approx \nabla f(x) + \nabla^2 f(x) v \\ &= 0 \end{aligned}$$

- d_{nt} is steepest descent direction at x in local Hessian norm

Measure of the proximity of x to $x^\star$

Newton decrement: $\lambda(x) = \left(\nabla f(x)^\top \nabla^2 f(x)^{-1} \nabla f(x) \right)^{1/2}$

→ gives an estimate of $f(x) - p^\star$, using quadratic approximation $\hat{f}$:

$$f(x) - \inf_y \hat{f}(y) = \frac{1}{2} \lambda(x)^2$$

- equal to the norm of the Newton step in the quadratic Hessian norm

$$\lambda(x) = \left(d_{\text{nt}}^\top \nabla^2 f(x) d_{\text{nt}} \right)^{1/2}$$

- directional derivative in the Newton direction: $\nabla f(x)^\top d_{\text{nt}} = -\lambda(x)^2$
- affine invariant (unlike $\|\nabla f(x)\|_2$)

Unconstrained Newton method

given starting point $x^{(0)} \in \text{dom } f$, tolerance $\epsilon > 0$,

repeat:

- ➊ Compute the Newton step $d_{\text{nt}}^{(k)}$ and decrement $\lambda(x^{(k)})$.
 - ➋ Stopping criterion: **quit** if $\lambda^2/2 \leq \epsilon$
 - ➌ Choose step size t by backtracking line search.
 - ➍ Update: $x^{(k+1)} = x^{(k)} + td_{\text{nt}}^{(k)}$
-

- descent method: for all k , $f(x^{(k+1)}) < f(x^{(k)})$
- affine invariant: Newton iterates for $\tilde{f}(y) = f(Qy)$ with starting point $y^{(0)} = Q^{-1}x^{(0)}$ are $y^{(k)} = Q^{-1}x^{(k)}$.

Change of coordinates in Newton method

$\min_{x \in \mathbb{R}^n} f(x)$ iterations: $x^{(k+1)} = x^{(k)} - \gamma (\nabla^2 f(x^{(k)}))^{-1} \nabla f(x^{(k)})$

- $x = Qy$ and $g(y) = f(Qy)$ ($Q \in \mathbb{R}^{n \times n}$ invertible)
- Iterations for equivalent problem $\min_{y \in \mathbb{R}^n} g(y)$:

$$\begin{aligned} y^{(k+1)} &= y^{(k)} - \gamma \left(\underbrace{\nabla^2 g(y^{(k)})}_{\substack{\text{Hessian of } g \\ \text{at } y^{(k)}}} \right)^{-1} \underbrace{\nabla g(y^{(k)})}_{\substack{\text{Gradient of } g \\ \text{at } y^{(k)}}} \\ &= y^{(k)} - \gamma \left(Q^\top \nabla^2 f(Qy^{(k)}) Q \right)^{-1} Q^\top \nabla f(Qy^{(k)}) \\ &= y^{(k)} - \gamma Q^{-1} \left(\nabla^2 f(Qy^{(k)}) \right)^{-1} \nabla f(Qy^{(k)}) \end{aligned}$$

Same iterations as original problem after multiplying by Q

➔ Newton method is **affine invariant**

Convergence of Newton algorithm

For f strongly convex ($\nabla^2 f(x) \succeq m\mathbf{Id}$) and Hessian L_H -Lipschitz, there exist η, γ with $0 < \eta \leq m^2/L_H$, $\gamma > 0$:

- if $\|\nabla f(x^{(k)})\|_2 \geq \eta$ (**damped phase**):

$$f(x^{(k+1)}) - f(x^{(k)}) \leq -\gamma$$

- if $\|\nabla f(x^{(k)})\|_2 < \eta$ (**quadratically convergent phase**), backtracking selects unit step and:

$$\frac{L_H}{2m^3} \|\nabla f(x^{(k+1)})\|_2 \leq \left(\frac{L_H}{2m^3} \|\nabla f(x^{(k)})\|_2 \right)^2$$

→ number of iterations until $f(x) - p^* \leq \epsilon$ bounded above by:

$$\frac{f(x^{(0)}) - p^*}{\gamma} + \log_2 \log_2 \left(\frac{2m^3}{L_H^2 \epsilon} \right)$$

Equality constrained minimization

With $f : \mathbb{R}^n \rightarrow \mathbb{R}$ convex, twice differentiable, find solution to:

$$\begin{cases} \min. f(x) \\ \text{s.t. } Ax = b \end{cases}$$

- Optimality condition: there exists a ν^* such that:

$$\begin{cases} Ax^* = b \\ \nabla f(x^*) + A^\top \nu^* = 0 \end{cases}$$

Equality constr. convex quad. minimization (example)

KKT conditions

$$\begin{cases} \min. \frac{1}{2}x^\top Px + q^\top x + r \\ \text{s.t. } Ax = b \end{cases} \quad \text{with } P \in \mathbb{S}_+^n.$$

- Lagrangian: $\mathcal{L}(x, \nu) = \frac{1}{2}x^\top Px + q^\top x + r + \nu^\top (Ax - b)$
- KKT conditions:

$$Ax^\star = b, \quad Px^\star + q + A^\top \nu^\star = 0$$

can be written as:

$$\begin{bmatrix} P & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} x^\star \\ \nu^\star \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}$$

Newton step at feasible point (equality constraints)

For x **feasible** ($Ax = b$), Newton step d_{nt} given by:

$$\begin{bmatrix} \nabla^2 f(x) & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} d_{\text{nt}} \\ w \end{bmatrix} = \begin{bmatrix} -\nabla f(x) \\ 0 \end{bmatrix}$$

Interpretation:

- minimizes second order approximation, i.e. finds small variation v :

$$\min_v \underbrace{f(x) + \nabla f(x)^\top v + \frac{1}{2} v^\top \nabla^2 f(x) v}_{\approx f(x+v)} \quad \text{s.t. } A(x+v) = b$$

- linearized optimality conditions.

$$\text{Newton decrement: } \lambda(x) = \left(d_{\text{nt}}^\top \nabla^2 f(x)^{-1} d_{\text{nt}} \right)^{1/2} = \left(-\nabla f(x)^\top d_{\text{nt}} \right)^{1/2}$$

Newton method with equality constraints and feasible point

given starting point $x^{(0)} \in \text{dom } f$ with $Ax^{(0)} = b$ (**feasible**),
tolerance $\epsilon > 0$,
repeat:

- ➊ Compute the Newton step $d_{\text{nt}}^{(k)}$ and decrement $\lambda(x^{(k)})$.
 - ➋ Stopping criterion: **quit** if $\lambda(x^{(k)})^2/2 \leq \epsilon$
 - ➌ Choose step size t by backtracking line search.
 - ➍ Update: $x^{(k+1)} = x^{(k)} + td_{\text{nt}}^{(k)}$
-

- **feasible** descent method: for all k , $f(x^{(k+1)}) < f(x^{(k)})$ and $x^{(k)}$ feasible
- affine invariant

Newton step at infeasible point (equality constraints)

For x **infeasible** ($Ax \neq b$), Newton step d_{nt} given by:

$$\begin{bmatrix} \nabla^2 f(x) & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} d_{\text{nt}} \\ w \end{bmatrix} = - \begin{bmatrix} \nabla f(x) \\ Ax - b \end{bmatrix}$$

Interpretation:

- minimizes second order approximation i.e. finds small variation v :

$$\min_v \underbrace{f(x) + \nabla f(x)^\top v + \frac{1}{2} v^\top \nabla^2 f(x) v}_{\approx f(x+v)} \quad \text{s.t. } A(x+v) = b$$

- linearized optimality conditions
- primal-dual** interpretation
 - write optimality conditions as $r(y) = 0$, where:

$$y = (x, \nu) \quad r(y) = (\nabla f(x) + A^\top \nu, Ax - b)$$

- linearizing $r(y) = 0$ gives $r(y + \Delta y) \approx r(y) + Dr(y)\Delta y = 0$ and yields the above equation with $w = \nu + \Delta \nu_{\text{nt}}$.

Infeasible start Newton method

given starting point $x^{(0)} \in \text{dom } f$, $\nu^{(0)}$,
 tolerance $\epsilon > 0$, $\alpha \in]0, 1/2[$, $\beta \in]0, 1[$
repeat:

- ➊ Compute primal and dual Newton steps $d_{\text{nt}}^{(k)}$, $\Delta\nu_{\text{nt}}^{(k)}$
- ➋ Backtracking line search on $\|r\|_2$.
 $t := 1$
while $\|r(x + td_{\text{nt}}^{(k)}, \nu + t\Delta\nu_{\text{nt}}^{(k)})\|_2 > (1 - \alpha t)\|r(x, \nu)\|_2$, $t := \beta t$
- ➌ Update: $x^{(k+1)} = x^{(k)} + td_{\text{nt}}^{(k)}$, $\nu^{(k+1)} = \nu^{(k)} + t\Delta\nu_{\text{nt}}^{(k)}$

until $Ax = b$ and $\|r(x, \nu)\|_2 \leq \epsilon$

- not a descent method: $f(x^{(k+1)}) > f(x^{(k)})$ is possible

Inequality constrained minimization

Notations and assumptions

$$p^* : \begin{cases} \min. f(x) \\ \text{s.t. } g_i(x) \leq 0, \\ \quad i = 1, \dots, m \\ Ax = b \end{cases} \quad \text{KKT cond.:} \begin{cases} Ax^* = b, \quad g_i(x^*) \leq 0 \\ \lambda^* \succeq 0 \\ \lambda_i^* g_i(x^*) = 0 \\ \nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(x^*) + A^\top \nu^* = 0 \end{cases}$$

Assumptions:

- functions f, g_i convex, twice continuously differentiable
- $A \in \mathbb{R}^{p \times n}$, $\text{rank } A = p$
- p^* is finite and attained
- problem is strictly feasible: there exist $\tilde{x}$ with
 $\tilde{x} \in \text{dom } f \quad g_i(\tilde{x}) < 0, i = 1, \dots, m, \quad A\tilde{x} = b$

→ strong duality holds, dual optimum is attained.

Inequality constrained minimization

Reformulation

$$p^* : \begin{cases} \min. f(x) \\ \text{s.t. } g_i(x) \leq 0, & i = 1, \dots, m \\ Ax = b \end{cases}$$

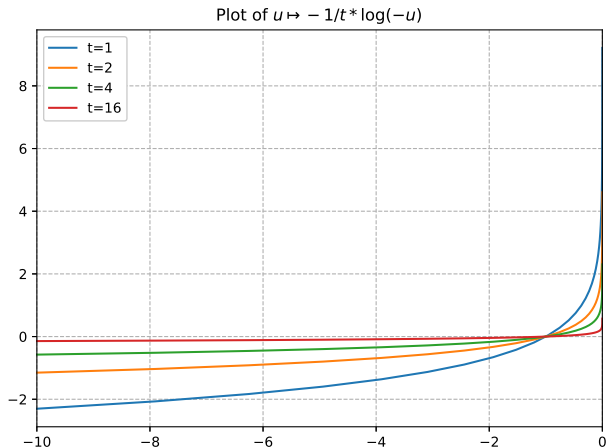
Equivalent equality constrained problem:

$$p^* : \min. f(x) + \sum_{i=1}^m v_{\mathbb{R}_-}(g_i(x)) \quad \text{s.t. } Ax = b$$

where $v_{\mathbb{R}_-}(u) = 0$ if $u \leq 0$ and $+\infty$ otherwise is **indicator function**.

→ Find an approximation for $v_{\mathbb{R}_-}$.

Logarithmic barrier



- For $t > 0$, $u \mapsto -\frac{1}{t} \log(-u)$ is a smooth approximation of $\iota_{\mathbb{R}_-}$
- Approximation improves as $t \rightarrow \infty$

Approximate problem and central path

$$p^* : \quad \min. f(x) + \sum_{i=1}^m \iota_{\mathbb{R}_-}(g_i(x)) \quad \text{s.t. } Ax = b$$

Approximation with **logarithmic barrier**:

$$\min. f(x) - \frac{1}{t} \sum_{i=1}^m \log(-g_i(x)) \quad \text{s.t. } Ax = b$$

Central path is $\{x^*(t) \mid t > 0\}$, where $x^*(t)$ is above optimum point.

$\rightsquigarrow$ can be solved by Newton method for each $t > 0$.

KKT of central path $\min. f(x) - \frac{1}{t} \sum_{i=1}^m \log(-g_i(x)) \quad \text{s.t. } Ax = b$

Optimal $(x^*(t), \nu^*(t))$ satisfy:

$$\begin{cases} Ax^*(t) = b, & g_i(x^*(t)) < 0 \\ \nabla f(x^*(t)) + \sum_{i=1}^m \frac{1}{-tg_i(x^*(t))} \nabla g_i(x^*(t)) + A^\top \nu^*(t) = 0 \end{cases}$$

KKT of central path $\min. f(x) - \frac{1}{t} \sum_{i=1}^m \log(-g_i(x)) \quad \text{s.t. } Ax = b$

Optimal $(x^*(t), \nu^*(t))$ satisfy:

$$\begin{cases} Ax^*(t) = b, & g_i(x^*(t)) < 0 \\ \nabla f(x^*(t)) + \sum_{i=1}^m \underbrace{\frac{1}{-tg_i(x^*(t))}}_{=: \lambda_i^*(t)} \nabla g_i(x^*(t)) + A^\top \nu^*(t) = 0 \end{cases}$$

$\rightarrow (x^*(t), \lambda^*(t), \nu^*(t))$ are solutions of **perturbed KKT**:

$$\begin{cases} Ax = b, & g_i(x) \leq 0 \\ \lambda \succeq 0 \\ -\lambda_i g_i(x) = \textcolor{red}{1/t} \\ \nabla f(x) + \sum_{i=1}^m \lambda_i \nabla g_i(x) + A^\top \nu = 0 \end{cases}$$

KKT of central path $\min_x f(x) - \frac{1}{t} \sum_{i=1}^m \log(-g_i(x)) \quad \text{s.t. } Ax = b$

Optimal $(x^*(t), \nu^*(t))$ satisfy:

$$\begin{cases} Ax^*(t) = b, & g_i(x^*(t)) < 0 \\ \nabla f(x^*(t)) + \sum_{i=1}^m \underbrace{\frac{1}{-tg_i(x^*(t))}}_{=: \lambda_i^*(t)} \nabla g_i(x^*(t)) + A^\top \nu^*(t) = 0 \end{cases}$$

$\rightarrow (\lambda^*(t), \nu^*(t))$ are dual feasible because - $\lambda_i^*(t) > 0$

- $x^*(t)$ minimizes $x \mapsto \mathcal{L}(x, \lambda^*(t), \nu^*(t))$

$$\begin{aligned} p^* &\geq \underbrace{\mathcal{L}_D(\lambda^*(t), \nu^*(t))}_{=\mathcal{L}(x^*(t), \lambda^*(t), \nu^*(t))} \\ &\geq f(x^*(t)) + \sum_{i=1}^m \lambda_i^*(t) g_i(x^*(t)) + \nu^*(t)^\top (Ax^*(t) - b) \\ &\geq f(x^*(t)) - \frac{m}{t} \end{aligned}$$

Barrier method

Given strictly feasible x , $t = t^{(0)}$, $\mu > 1$, tolerance $\epsilon > 0$,
repeat:

- ➊ Centering step. Compute $x^*(t)$ (using equality constrained Newton).
 - ➋ Update. $x := x^*(t)$.
 - ➌ Stopping criterion. **quit** if $m/t \leq \epsilon$.
 - ➍ Increase t . $t := \mu t$.
-

- Terminates with $f(x) - p^* \leq \epsilon$
- Centering usually done using Newton's method, starting at current x
- Choice of μ involves a trade-off: large μ means fewer outer iterations, more inner (Newton) iterations; typical values: $\mu = 10 - 20$.
- Several heuristics for choice of $t^{(0)}$

Feasibility and phase I methods

Feasibility problem: find x such that

$$g_i(x) \leq 0, \quad i = 1, \dots, m, \quad Ax = b \quad (1)$$

Phase I: computes strictly feasible point for barrier method

Basic phase I method

$$\begin{cases} \min. s \\ \text{s.t. } g_i(x) \leq s, \quad i = 1, \dots, m \\ Ax = b \end{cases} \quad (2)$$

- If x, s feasible with $s < 0$, then x strictly feasible for (1).
- If optimal value $\bar{p}^*$ of (2) is positive, then (1) infeasible.
- If $\bar{p}^* = 0$ in (2) and attained, then (1) feasible (but not strictly).
if $\bar{p}^* = 0$ in (2) and not attained, then (1) infeasible.

Generalized inequalities

$$\begin{cases} \min. f(x) \\ \text{s.t. } g_i(x) \prec_{K_i} 0, & i = 1, \dots, m \\ Ax = b \end{cases}$$

- f convex
- $g_i : \mathbb{R}^n \rightarrow \mathbb{R}^{k_i}$ convex with respect to proper cones $K_i \subset \mathbb{R}^{k_i}$
- f, g_i twice continuously differentiable
- $A \in \mathbb{R}^{p \times n}$ with $\text{rank} A = p$
- We assume p^* is finite and attained
- We assume problem is strictly feasible; hence strong duality holds and dual optimum is attained

↪ Ex: SOCP, SDP

(A few words about) Convergence

Number of outer (centering) iterations: exactly

$$\left\lceil \frac{\log(m/(\epsilon t^{(0)}))}{\log \mu} \right\rceil$$

plus the initial centering step (to compute $x^*(t^{(0)})$)

Centering problem: see convergence analysis of Newton's method

Part V

Proximal methods

Generalities about proximal methods

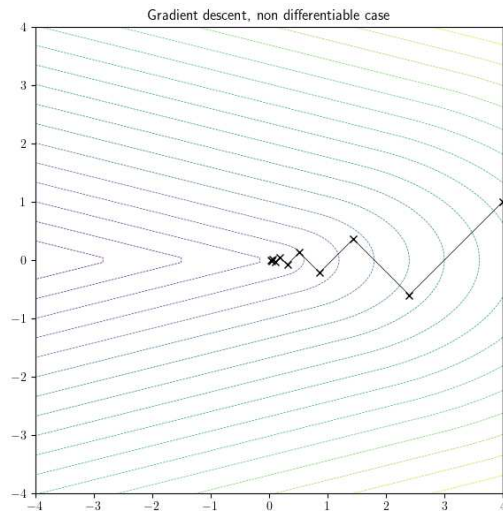
Gradient and Newton methods:

- smooth functions (differentiable once or twice),
- medium size problems (Newton), sometimes larger (gradient)

Proximal methods:

- suitable for **smooth** and **non-smooth** functions,
- suitable for **constrained and unconstrained** problems,
- large size and distributed implementations,
- based on high level "prox" operation, which is itself an optimization problem.

(Sub)-gradient in non differentiable case



$$f(x, y) = \begin{cases} \sqrt{x^2 + \eta y^2} & \text{if } |y| \leq x, \\ \frac{x + \eta|y|}{\sqrt{1 + \eta}} & \text{if } |y| \geq x. \end{cases}$$

Optimal step size

Starting point: $(x^{(0)}, y^{(0)}) = (\eta, 1)$

Proximal operator

Let f be a **closed proper convex** function.

Proximal operator

$$\text{prox}_f(v) = \text{Arg min}_x \left[f(x) + \frac{1}{2} \|x - v\|_2^2 \right]$$

Proximal operator of the **scaled** function (with $\lambda > 0$)

$$\text{prox}_{\lambda f}(v) = \text{Arg min}_x \left[f(x) + \frac{1}{2\lambda} \|x - v\|_2^2 \right]$$

Projection and prox

With ι_C indicator function of convex set C , **proximal operator generalizes projection** Π_C :

$$\begin{aligned}\text{prox}_{\lambda \iota_C}(v) &= \text{Arg min}_x \left[\iota_C(x) + \frac{1}{2\lambda} \|x - v\|_2^2 \right] \\ &= \text{Arg min}_{x \in C} \|x - v\|_2^2 \\ &= \Pi_C(v)\end{aligned}$$

- Ex: for C an affine subset $C = \{x \mid Ax = b\}$:

$$\text{prox}_{\iota_{\{x \mid Ax=b\}}}(v) = (\mathbf{Id} - A^\top (AA^\top)^{-1} A)v + A^\top (AA^\top)^{-1} b$$

Prox: examples

Affine function: $f(x) = b^\top x + c$:

$$\text{prox}_{\lambda f}(v) = v - \lambda b$$

Quadratic function: $f(x) = \frac{1}{2}x^\top Ax + b^\top x + c$ with $A \in \mathbb{S}_+^n$

$$\text{prox}_{\lambda f}(v) = (\mathbf{Id} + \lambda A)^{-1}(v - \lambda b)$$

Indeed: above expression(s) obtained by setting derivative to zero

$$\nabla f(x) + \frac{1}{\lambda}(x - v) = Ax + b + \frac{1}{\lambda}(x - v) = 0$$

- **Shrinkage operator:** $\text{prox}_{\frac{\lambda}{2}(\cdot)^2}(v) = \frac{1}{1+\lambda}v$ or more generally:

$$\text{prox}_{\frac{\lambda}{2}\|\cdot\|_2^2}(v) = \frac{1}{1+\lambda}v$$

- For 1st order approximation $\hat{f}_1(x) = f(x_0) + \nabla f(x_0)^\top (x - x_0)$:

$$\text{prox}_{\lambda \hat{f}_1}(x_0) = x_0 - \lambda \nabla f(x_0)$$

- For 2nd order approximation

$$\hat{f}_2(x) = f(x_0) + \nabla f(x_0)^\top (x - x_0) + \frac{1}{2}(x - x_0)^\top \nabla^2 f(x_0)(x - x_0):$$

$$\text{prox}_{\lambda \hat{f}_2}(x_0) = x_0 - \left(\frac{1}{\lambda} \mathbf{Id} + \nabla^2 f(x_0) \right)^{-1} \nabla f(x_0)$$

Interpretation of prox

$$\text{prox}_{\lambda f}(v) = \text{Arg min}_x \left[f(x) + \frac{1}{2\lambda} \right] \|x - v\|_2^2$$

- $\text{prox}_{\lambda f}(v)$ moves from v towards the minimum of f , penalized by the cost of staying near to v depending on λ
- Connection with gradient step (under some assumptions, for small λ):

$$\text{prox}_{\lambda f}(v) \approx v - \lambda \nabla f(v)$$

Prox and subdifferential

From $\text{prox}_{\lambda f}(v) = \text{Arg min}_x [f(x) + \frac{1}{2\lambda}\|x - v\|_2^2]$, it follows:

$$\begin{aligned} p = \text{prox}_{\lambda f}(v) &\Leftrightarrow 0 \in \partial f(p) + \frac{1}{\lambda}(p - v) \\ &\Leftrightarrow v \in p + \lambda \partial f(p) \\ &\Leftrightarrow v \in (\mathbf{Id} + \lambda \partial f)(p) \end{aligned}$$

Resolvent

For an operator T , the resolvent of T is $(\mathbf{Id} + \lambda T)^{-1}$.

Resolvent of subdifferential

$$\text{prox}_{\lambda f} = (\mathbf{Id} + \lambda \partial f)^{-1}$$

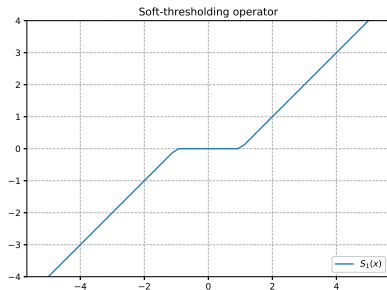
In addition, $\text{prox}_{\lambda f}$ is single-valued.

Soft thresholding

(Scalar case)

$\text{prox}_{\lambda|\cdot|}(\cdot)$ of absolute value is the **soft thresholding** operator:

$$S_{\lambda}(v) = \text{sign}(v) [|v| - \lambda]_+ = \begin{cases} v - \lambda & \text{if } v \geq \lambda, \\ 0 & \text{if } -\lambda \leq v \leq \lambda, \\ v + \lambda & \text{if } v \leq -\lambda. \end{cases}$$



prox of separable sum

If $f(x) = \sum_{i=1}^n f_i(x_i)$,

$$\text{prox}_f(v) = \begin{bmatrix} \text{prox}_{f_1}(v_1) \\ \vdots \\ \text{prox}_{f_n}(v_n) \end{bmatrix}$$

► For $f(x) = \|x\|_1$:

$$\left[\text{prox}_{\lambda \|\cdot\|_1}(v) \right]_i = S_\lambda(v_i)$$

► For $f(x) = \frac{1}{2}\|x\|_2^2$:

$$\text{prox}_{\frac{\lambda}{2}\|\cdot\|_2^2}(v) = \left(\frac{1}{1+\lambda} \right) v$$

Other properties of prox

- Precomposition: if $\tilde{f}(x) = f(\alpha x + \beta)$,

$$\text{prox}_{\lambda \tilde{f}}(v) = \frac{1}{\alpha} [\text{prox}_{\alpha^2 \lambda f}(\alpha v + \beta) - \beta]$$

- Postcomposition: if $\tilde{f}(x) = \alpha f(x) + b$ with $\alpha > 0$,

$$\text{prox}_{\lambda \tilde{f}}(v) = \text{prox}_{\alpha \lambda f}(v)$$

- Affine addition: if $\tilde{f}(x) = f(x) + a^\top x + b$,

$$\text{prox}_{\lambda \tilde{f}}(v) = \text{prox}_{\lambda f}(v - \lambda a)$$

- Regularization: if $\tilde{f}(x) = f(x) + \rho/2 \|x - a\|_2^2$,

$$\text{prox}_{\lambda \tilde{f}}(v) = \text{prox}_{\tilde{\lambda} f}((\tilde{\lambda}/\lambda)v + (\rho \tilde{\lambda})a) \text{ where } \tilde{\lambda} = \lambda/(1 + \lambda\rho)$$

Moreau decomposition

Let $f^*(v) = \sup_x \langle v, x \rangle - f(x)$ be the Fenchel conjugate of f .

Moreau decomposition

$$v = \text{prox}_f(v) + \text{prox}_{f^*}(v)$$

Moreau decomposition

Let $f^*(v) = \sup_x \langle v, x \rangle - f(x)$ be the Fenchel conjugate of f .

Moreau decomposition

$$v = \text{prox}_f(v) + \text{prox}_{f^*}(v)$$

Proof: Let $p = \text{prox}_f(v)$ and define $q = v - p$. By definition of prox , $q \in \partial f(p)$ and hence $p \in \partial f^*(q)$, which means $v - q \in \partial f^*(q)$ and hence $q = \text{prox}_{f^*}(v)$.

Moreau decomposition

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Proof: Let $p = \text{prox}_f(v)$ and define $q = v - p$. By definition of prox , $q \in \partial f(p)$ and hence $p \in \partial f^*(q)$, which means $v - q \in \partial f^*(q)$ and hence $q = \text{prox}_{f^*}(v)$.

► generalizes orthogonal decomposition:

► take L a subspace and $f = \iota_L$:

$$\begin{aligned} \iota_L^*(v) &= \sup_x (v^\top x - \iota_L(x)) = \sup_{x \in L} v^\top x \\ &= \begin{cases} +\infty & \text{if } v^\top x_0 \neq 0 \text{ for an } x_0 \in L \\ 0 & \text{if } v^\top x = 0 \text{ for all } x \in L \end{cases} = \iota_{L^\perp}(v) \\ &\text{where } L^\perp = \{y \mid y^\top x = 0 \text{ for all } x \in L\} \end{aligned}$$

► The Moreau decomposition reads: $v = \Pi_L(v) + \Pi_{L^\perp}(v)$

Fixed points of prox

Minimizers of f are fixed points of prox_f :

$$x^* \text{ minimizes } f \Leftrightarrow x^* = \text{prox}_f(x^*)$$

Proof:

- $\Rightarrow$ $f(x) \geq f(x^*)$ for any x hence $f(x) + \frac{1}{2}\|x - x^*\|_2^2 \geq f(x^*) + \frac{1}{2}\|x^* - x^*\|_2^2$
which proves that x^* minimizes the l.h.s. expression.
- $\Leftarrow$ $\tilde{x} = \text{prox}_f(v)$ if and only if $\tilde{x}$ minimizes $f(x) + \frac{1}{2}\|x - v\|_2^2$, that is if and only if $0 \in \partial f(\tilde{x}) + (\tilde{x} - v)$. With $\tilde{x} = v$, we get $0 \in \partial f(\tilde{x})$ and thus $\tilde{x} = v = x^*$.

Proximal point algorithm

Proximal minimization algorithm

$$x^{(k+1)} = \text{prox}_{\lambda f} \left(x^{(k)} \right)$$

- Convergence can be justified, few applications.
- Iterative refinement method for solving $Ax = b$ ($A \in \mathbb{S}_+^n$):

$$x^{(k+1)} = x^{(k)} + (A + \epsilon \mathbf{Id})^{-1}(b - Ax^k)$$

↔ Proximal point minimization of $g(x) = \frac{1}{2}x^\top Ax - b^\top x$:

$$\begin{aligned} \text{prox}_{\lambda g}(v) &= (\mathbf{Id} + \lambda A)^{-1}(v + \lambda Av - \lambda Av + \lambda b) \\ &= v - \left(\frac{1}{\lambda} \mathbf{Id} + A\right)^{-1}(Av - b) \end{aligned}$$

Proximal gradient

- Split objective:

$$\min. f(x) + g(x)$$

$f : \mathbb{R}^n \rightarrow \mathbb{R}$, $g : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{\infty\}$ are l.s.c., proper, convex;
 f is **differentiable** and g can be **nonsmooth**

- Proximal gradient method:

$$x^{(k+1)} := \text{prox}_{\lambda_k g} \left(x^{(k)} - \lambda_k \nabla f(x^{(k)}) \right)$$

where $\lambda_k > 0$ is a step size.

- ▷ Converges with fixed step size $\lambda_k = \lambda \in]0, 2/L[$ when ∇f is Lipschitz continuous with constant L .

LASSO (Least Absolute Shrinkage and Selection Operator)

(Proximal gradient algorithm)

$$\min. \frac{1}{2} \|Ax - b\|_2^2 + \gamma \|x\|_1$$

- Splitting:

$$f(x) = \frac{1}{2} \|Ax - b\|_2^2$$

$$g(x) = \gamma \|x\|_1$$

$$\nabla f(x) = A^\top (Ax - b)$$

$$\text{prox}_{\lambda g}(x) = S_{\lambda\gamma}(x)$$

- Proximal algorithm:

$$x^{(k+1)} := S_{\lambda\gamma} \left(x^{(k)} - \lambda A^\top (Ax^{(k)} - b) \right)$$

where fixed step-size $0 < \lambda \leq \frac{1}{\|A^\top A\|_2}$

- ▶ Sometimes called ISTA (Iterative Shrinkage-Thresholding Algorithm), accelerated version called FISTA (Fast ISTA).

Alternating Direction Method of Multipliers (ADMM)

(seen as a proximal algorithm)

- Split objective:

$$\min. f(x) + g(x)$$

$f, g : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{\infty\}$ are l.s.c., proper, convex.

f and g can be **nonsmooth**.

- Alternating direction method of multipliers (**ADMM**):

$$\begin{cases} x^{(k+1)} := \text{prox}_{\lambda f}(z^{(k)} - u^{(k)}) \\ z^{(k+1)} := \text{prox}_{\lambda g}(x^{(k+1)} + u^{(k)}) \\ u^{(k+1)} := u^{(k)} + x^{(k+1)} - z^{(k+1)} \end{cases}$$

▷ Also known as Douglas-Rachford splitting.

Augmented Lagrangian and prox operator

- min. $f(x) + g(x)$ equivalent to:

$$\begin{cases} \min. f(x) + g(z) \\ \text{s.t. } x - z = 0 \end{cases}$$

- Augmented Lagrangian (with parameter $\rho > 0$):

$$L_\rho(x, z, y) = f(x) + g(z) + y^\top(x - z) + \frac{\rho}{2}\|x - z\|_2^2$$

can be written with $u = \frac{1}{\rho}y$:

$$L_\rho(x, z, y) = f(x) + g(z) + \frac{\rho}{2}\|x - z + u\|_2^2 - \frac{\rho}{2}\|u\|_2^2$$

$$\Rightarrow \operatorname{Arg} \min_x L_\rho(x, z, y) = \operatorname{prox}_{\lambda f}(z - u)$$

$$\Rightarrow \operatorname{Arg} \min_z L_\rho(x, z, y) = \operatorname{prox}_{\lambda g}(x + u) \text{ where } \lambda = \frac{1}{\rho}.$$

Alternating Direction Method of Multipliers (ADMM)

(seen as an augmented Lagrangian method)

- $\min. f(x) + g(x)$ equivalent to:

$$\begin{cases} \min. f(x) + g(z) \\ \text{s.t. } x - z = 0 \end{cases}$$

- Augmented Lagrangian (with parameter $\rho > 0$):

$$L_\rho(x, z, y) = f(x) + g(z) + y^\top (x - z) + \frac{\rho}{2} \|x - z\|_2^2$$

- Alternate Direction Method of Multipliers (**ADMM**) iterations:

$$\begin{cases} x^{(k+1)} := \text{Arg min}_x L_\rho(x, z^{(k)}, y^{(k)}) \\ z^{(k+1)} := \text{Arg min}_z L_\rho(x^{(k+1)}, z, y^{(k)}) \\ y^{(k+1)} := y^{(k)} + \rho(x^{(k+1)} - z^{(k+1)}) \end{cases}$$

Basis pursuit

(ADMM algorithm)

$$\begin{cases} \min. \|x\|_1 \\ \text{s.t. } Ax = b \end{cases}$$

- Equivalent to:

$$\begin{cases} \min. \iota_{\{x \mid Ax=b\}}(x) + \|z\|_1 \\ \text{s.t. } x - z = 0 \end{cases}$$

- ADMM iterations (derived from slide 147):

$$\begin{cases} x^{(k+1)} := \Pi_{\{x \mid Ax=b\}}(z^{(k)} - u^{(k)}) \\ z^{(k+1)} := S_\lambda(x^{(k+1)} + u^{(k)}) \\ u^{(k+1)} := u^{(k)} + x^{(k+1)} - z^{(k+1)} \end{cases}$$

with S_λ : a soft thresholding and $\Pi_{\{x \mid Ax=b\}}$: projection.

LASSO (Least Absolute Shrinkage and Selection Operator)

(ADMM algorithm)

$$\min. \frac{1}{2} \|Ax - b\|_2^2 + \gamma \|x\|_1$$

- Equivalent to:

$$\begin{cases} \min. \frac{1}{2} \|Ax - b\|_2^2 + \gamma \|z\|_1 \\ \text{s.t. } x - z = 0 \end{cases}$$

- ADMM iterations (derived from slide 147):

$$\begin{cases} x^{(k+1)} := (\lambda A^\top A + \mathbf{Id})^{-1} ((z^{(k)} - u^{(k)}) + \lambda A^\top b) \\ z^{(k+1)} := S_{\lambda\gamma}(x^{(k+1)} + u^{(k)}) \\ u^{(k+1)} := u^{(k)} + x^{(k+1)} - z^{(k+1)} \end{cases}$$

with $S_{\lambda\gamma}$: soft thresholding.

Alternating Direction Method of Multipliers (ADMM)

(seen as an augmented Lagrangian method)

$$\begin{cases} \min. f(x) + g(z) \\ \text{s.t. } Ax + Bz = c \end{cases}$$

- Augmented Lagrangian (with parameter $\rho > 0$):

$$L_\rho(x, z, y) = f(x) + g(z) + y^\top (Ax + Bz - c) + \frac{\rho}{2} \|Ax + Bz - c\|_2^2$$

- Alternate Direction Method of Multipliers (**ADMM**) iterations:

$$\begin{cases} x^{(k+1)} := \text{Arg min}_x L_\rho(x, z^{(k)}, y^{(k)}) \\ z^{(k+1)} := \text{Arg min}_z L_\rho(x^{(k+1)}, z, y^{(k)}) \\ y^{(k+1)} := y^{(k)} + \rho(Ax^{(k+1)} + Bz^{(k+1)} - c) \end{cases}$$

Generalized LASSO

(ADMM algorithm)

$$\min. \frac{1}{2} \|Ax - b\|_2^2 + \gamma \|Fx\|_1$$

- Equivalent to:

$$\begin{cases} \min. \frac{1}{2} \|Ax - b\|_2^2 + \gamma \|z\|_1 \\ \text{s.t. } Fx - z = 0 \end{cases}$$

- ADMM iterations (derived from slide 152 with $\rho = 1/\lambda$, compare with slide 151):

$$\begin{cases} x^{(k+1)} := (A^\top A + \rho F^\top F)^{-1} (A^\top b + \rho F^\top (z^{(k)} - u^{(k)})) \\ z^{(k+1)} := S_{\gamma/\rho}(Fx^{(k+1)} + u^{(k)}) \\ u^{(k+1)} := u^{(k)} + Fx^{(k+1)} - z^{(k+1)} \end{cases}$$

Part VI

Appendix

Some notations

$\mathbb{R}$	real numbers
$\mathbb{R}_+$	nonnegative (≥ 0) numbers
$\mathbb{R}_{++}$	positive (> 0) numbers
$\mathbb{S}^n$	$n \times n$ real-valued symmetric matrices
$\mathbb{S}_+^n / \mathbb{S}_{++}^n$	$n \times n$ sym. positive semidefinite /definite matrices
$\text{tr } A$	trace of the matrix A
$A^\top$	transpose of the matrix A
$\mathbf{1} / \mathbf{1}_n$	all ones (column) vector / all ones vector of size n
$\ \cdot\ _2$	Euclidian norm
$\ \cdot\ _1 / \ \cdot\ _\infty$	ℓ_1 / ℓ_∞ norm
$\sup / \inf$	supremum / infimum
$\preceq_K / \succeq_K / \prec_K / \succ_K$ $\preceq / \succeq (\prec / \succ)$	inequalities wrt to cone K . If not specified, K is positive orthant or $\mathbb{S}_+^n$
$[\cdot]_+$	positive part $[x]_+ = \max(0, x)$

Euclidian space

Euclidian space $\mathbf{E}$ (finite dimension) with inner-product $\langle \cdot, \cdot \rangle$

- Often $\mathbf{E} = \mathbb{R}^n$ and $\langle x, y \rangle = x^\top y = \sum_{i=1}^n x_i y_i$
- (Euclidian) norm $\|x\|_2 = \sqrt{\langle x, x \rangle}$
- Cauchy-Schwarz inequality: $|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2$
- Orthogonal complement:

$$G^\perp = \{y \in \mathbf{E} \mid \langle x, y \rangle = 0 \text{ for all } x \in G\}$$

- Ball of center x_0 radius $r \geq 0$:

$$B(x_0, r] = \{x \in \mathbf{E} \mid \|x - x_0\| \leq r\} \quad (\text{closed ball})$$

$$B(x_0, r[= \{x \in \mathbf{E} \mid \|x - x_0\| < r\} \quad (\text{open ball})$$

Linear maps, null space, adjoint

$\mathbf{E}$ and $\mathbf{F}$ two Euclidian spaces.

- $A : \mathbf{E} \rightarrow \mathbf{F}$ is linear if $A(\lambda x + \mu y) = \lambda Ax + \mu Ay$ for any $x, y \in \mathbf{E}$ and $\lambda, \mu \in \mathbb{R}$.
- Null space (kernel): $\text{Ker } A = \{x \in \mathbf{E} \mid Ax = 0\}$
- Linear functions $\mathbf{E} \rightarrow \mathbb{R}$ have the form $\langle a, \cdot \rangle$ for some $a \in \mathbf{E}$
- **Affine** functions = linear + constant
- **Adjoint** of A is the linear map $A^* : \mathbf{F} \rightarrow \mathbf{E}$ such that:

$$\langle A^*y, x \rangle = \langle y, Ax \rangle \text{ for any } x \in \mathbf{E}, y \in \mathbf{F}$$

- ▶ If $\mathbf{E} = \mathbb{R}^n$, $\mathbf{F} = \mathbb{R}^p$, adjoint of $A : \mathbb{R}^n \rightarrow \mathbb{R}^p$ is given by $A^\top$

Dual norm

Let $\|\cdot\|$ be a norm on $\mathbf{E}$.

Associated **dual norm** $\|\cdot\|_*$

$$\|z\|_* := \sup_{\|x\| \leq 1} \langle z, x \rangle$$

- $\langle z, x \rangle \leq \|z\|_* \|x\|$
- Dual norm of $\|\cdot\|_2$ is itself.
- $\|\cdot\|_\infty$ and $\|\cdot\|_1$ are dual norms of each other.
- Dual of ℓ_p -norm is ℓ_q norm with $\frac{1}{p} + \frac{1}{q} = 1$.
- $\|\cdot\|_{**} = \|\cdot\|$ (need not hold in infinite dimensional spaces)

Open and closed sets

Interior, closure, boundary

interior of a set C :

$$\text{int } C = \{x \in C \mid B(x, \varepsilon) \subset C \text{ for sufficiently small } \varepsilon\}$$

A set C is **open** if $C = \text{int } C$ and **closed** if its complement is open.

closure of a set C :

$$\text{cl } C = \{x \in \mathbf{E} \mid \text{for any (small) } \varepsilon, B(x, \varepsilon) \cap C \neq \emptyset\}$$

boundary of a set C : $\text{bd } C = \text{cl } C \setminus \text{int } C$

core of a set C = set of points $x \in C$ such that for any direction $d \in \mathbf{E}$, $x + td \in C$ for all small t . Note that $\text{int } C \subseteq \text{core } C$ (but $\text{core } C$ may be larger than $\text{int } C$).

Symmetric matrices

- Set of symmetric matrices: $\mathbb{S}^n = \{M \in \mathbb{R}^{n \times n} \mid M^\top = M\}$
- Positive semidefinite matrices: $\mathbb{S}_+^n = \{M \in \mathbb{S}^n \mid x^\top M x \geq 0 \text{ for all } x\}$
- Positive definite matrices: $\mathbb{S}_{++}^n = \{M \in \mathbb{S}^n \mid x^\top M x > 0 \text{ for all } x \neq 0\}$
- Inner product:

$$\langle A, B \rangle = \text{tr } AB \text{ for } A, B \in \mathbb{S}^n$$

- $M \in \mathbb{S}_+^n$ (resp. $\mathbb{S}_{++}^n$) will be written $M \succeq 0$ (resp. $M \succ 0$).
Similarly (see later):

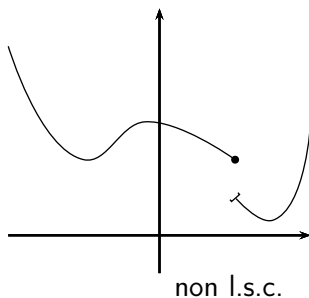
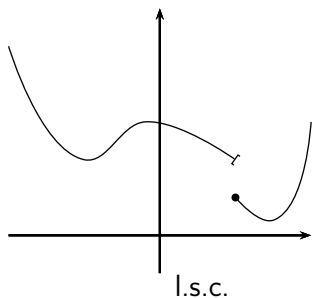
$$A - B \in \mathbb{S}_+^n \Leftrightarrow A \succeq B \qquad A - B \in \mathbb{S}_{++}^n \Leftrightarrow A \succ B$$

Lower semi-continuous function (l.s.c.)

f is l.s.c. if and only if at any point x :

$$x_n \xrightarrow{n \rightarrow \infty} x \Rightarrow f(x) \leq \lim_{n \rightarrow \infty} f(x_n)$$

f is l.s.c. $\Leftrightarrow$ epigraph $\{(x, t) \in \mathbb{R}^n \times \mathbb{R} \mid f(x) \leq t\}$ is a closed set



Remark: sometimes, l.s.c. function = closed function

Gradient Lipschitz function

f is **gradient Lipschitz** (with constant L) when:

$$\|\nabla f(x) - \nabla f(y)\|_2 \leq L\|x - y\|_2 \quad (\forall x, y)$$

For twice differentiable f , equivalent to $\nabla^2 f(x) \preceq L\mathbf{Id}$.

Consequences:

→ f is L -**strongly smooth**: $\frac{L}{2}\|\cdot\|_2^2 - f(\cdot)$ is convex

→ $f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|_2^2 \quad (\forall x, y)$

Strongly convex function

f is μ -**strongly convex** ($\mu > 0$) when $f(\cdot) - \frac{\mu}{2}\|\cdot\|_2^2$ is convex.

For twice differentiable f , equivalent to $\mu\mathbf{Id} \preceq \nabla^2 f(x)$.

Consequences:

$$\rightarrow f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2}\|y - x\|_2^2 \quad (\forall x, y) \text{ (convexity)}$$

$$\rightarrow p^* \geq f(x) - \frac{1}{2\mu}\|\nabla f(x)\|_2^2 \text{ (minimize above r.h.s. w.r.t. } y)$$

This is **PL-condition** (Polyak-Lojasiewicz) $\|\nabla f(x)\|_2^2 \geq 2\mu(f(x) - p^*)$

$\rightarrow$ For strongly convex f , sublevel sets are bounded. If twice continuously differentiable, upper-bound on Hessian exists: $\mu\mathbf{Id} \preceq \nabla^2 f(x) \preceq L\mathbf{Id}$